

Unit 1 AP Topics 9-12

1.

x	-1.5	-1.0	-0.5	0	0.5	1.0	1.5
$f(x)$	-1	-4	-6	-7	-6	-4	-1
$f'(x)$	-7	-5	-3	0	3	5	7

Let f be a function that is differentiable for all real numbers. The table above gives the values of f and its derivative f' for selected points x in the closed interval $-1.5 \leq x \leq 1.5$. The second derivative of f has the property that $f''(x) > 0$ for $-1.5 \leq x \leq 1.5$.

(a) Evaluate $\int_0^{1.5} (3f'(x) + 4)dx$. Show the work that leads to your answer.

(b) Write an equation of the line tangent to the graph of f at the point where $x = 1$. Use this line to approximate the value of $f(1.2)$. Is this approximation greater than or less than the actual value of $f(1.2)$? Give a reason for your answer.

(c) Find a positive real number r having the property that there must exist a value c with $0 < c < 0.5$ and $f'(c) = r$. Give a reason for your answer.

(d) Let g be the function given by $g(x) = \begin{cases} 2x^2 - x - 7 & \text{for } x < 0 \\ 2x^2 + x - 7 & \text{for } x \geq 0. \end{cases}$

The graph of g passes through each of the points $(x, f(x))$ given in the table above. Is it possible that f and g are the same function? Give a reason for your answer.

Part A

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned} \int_0^{1.5} (3f'(x) + 4)dx &= 3 \int_0^{1.5} f'(x)dx + \int_0^{1.5} 4dx \\ &= 3f(x) + 4x \Big|_0^{1.5} = 3(-1 - (-7)) + 4(1.5) = 24 \end{aligned}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the antiderivative

1 point is earned for the answer

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$$\begin{aligned}\int_0^{1.5} (3f'(x) + 4) dx &= 3 \int_0^{1.5} f'(x) dx + \int_0^{1.5} 4 dx \\ &= 3f(x) + 4x \Big|_0^{1.5} = 3(-1 - (-7)) + 4(1.5) = 24\end{aligned}$$

Part B

1 point is earned for the tangent line

1 point is earned for computing y on tangent line at $x = 1.2$

1 point is earned for the answer with reason

$$\begin{aligned}y &= 5(x - 1) - 4 \\ f(1.2) &\approx 5(0.2) - 4 = -3\end{aligned}$$

The approximation is less than $f(1.2)$ because the graph of f is concave up on the interval

$$1 < x < 1.2.$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the tangent line

1 point is earned for computing y on tangent line at $x = 1.2$

1 point is earned for the answer with reason

$$\begin{aligned}y &= 5(x - 1) - 4 \\ f(1.2) &\approx 5(0.2) - 4 = -3\end{aligned}$$

The approximation is less than $f(1.2)$ because the graph of f is concave up on the interval

$$1 < x < 1.2.$$

Part C

1 point is earned for the reference to MVT for f' (or differentiability of f)

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1 point is earned for the value of r for interval $0 \leq x \leq 0.5$

By the Mean Value Theorem there is a c with $0 < c < 0.5$ such that

$$f''(c) = \frac{f'(0.5) - f'(0)}{0.5 - 0} = \frac{3 - 0}{0.5} = 6 = r$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the reference to MVT for f' (or differentiability of f)

1 point is earned for the value of r for interval $0 \leq x \leq 0.5$

By the Mean Value Theorem there is a c with $0 < c < 0.5$ such that

$$f''(c) = \frac{f'(0.5) - f'(0)}{0.5 - 0} = \frac{3 - 0}{0.5} = 6 = r$$

Part D

1 point is earned for the answers "no" with reference to g' or g''

1 point is earned for the correct reason

$$\lim_{x \rightarrow 0^-} g'(x) = \lim_{x \rightarrow 0^-} (4x - 1) = -1$$

$$\lim_{x \rightarrow 0^+} g'(x) = \lim_{x \rightarrow 0^+} (4x + 1) = +1$$

Thus g' is not continuous at $x = 0$, but f' is continuous at $x = 0$, so $f \neq g$.

OR

$g''(x) = 4$ for all $x \neq 0$, but it was shown in part (c) that $f''(c) = 6$ for some $c \neq 0$, so $f \neq g$.

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0	1	2
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The student response earns all of the following points:

1 point is earned for the answers "no" with reference to g' or g''

1 point is earned for the correct reason

$$\lim_{x \rightarrow 0^-} g'(x) = \lim_{x \rightarrow 0^-} (4x - 1) = -1$$

$$\lim_{x \rightarrow 0^+} g'(x) = \lim_{x \rightarrow 0^+} (4x + 1) = +1$$

Thus g' is not continuous at $x = 0$, but f' is continuous at $x = 0$, so $f \neq g$.

OR

$g''(x) = 4$ for all $x \neq 0$, but it was shown in part (c) that $f''(c) = 6$ for some $c \neq 0$, so $f \neq g$.

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2. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be the function defined by $f(x) = \frac{ax^2+bx+2}{2x^2-8}$, where a and b are constants. The graph of f has a horizontal asymptote at $y = 3$, and f has a removable discontinuity at $x = 2$.

- Show that $a = 6$ and $b = -13$.
- To make f continuous at $x = 2$, $f(2)$ should be defined as what value? Justify your answer.
- At what value of x does f have a discontinuity due to a vertical asymptote? Give a reason for your answer.

Part A

The last point is earned for work that concludes with the correct values for both a, and b.

At most 3 out of 4 points if mathematical notation for limits is missing or incorrect.

Students do not have to show the factorization to earn the point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
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The student response accurately includes all four of the criteria below.

- ☐ recognizes $\lim_{x \rightarrow \infty} f(x) = 3$
- ☐ shows that $\lim_{x \rightarrow \infty} f(x) = \frac{a}{2}$
- ☐ sets $ax^2 + bx + 2 = 0$ at $x = 2$
- ☐ verification

Solution:

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$$\lim_{x \rightarrow \infty} f(x) = 3$$

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{x^2 \left(a + \frac{b}{x} + \frac{2}{x^2} \right)}{x^2 \left(2 - \frac{8}{x^2} \right)} = \lim_{x \rightarrow \infty} \frac{a + \frac{b}{x} + \frac{2}{x^2}}{2 - \frac{8}{x^2}} = \frac{a}{2}$$

$$\Rightarrow a = 6$$

Since f has a removable discontinuity at $x = 2$ and the denominator $2x^2 - 8$ equals 0 at $x = 2$, the numerator must also equal 0 at $x = 2$.

$$0 = 6 \cdot 2^2 + 2b + 2 = 26 + 2b$$

$$\Rightarrow b = -13$$

Part B

At most 1 out of 2 points if mathematical notation for any limit is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ uses $\lim_{x \rightarrow 2} f(x)$
- ☐ answer with justification

Solution:

f will be continuous at $x = 2$ if $f(2) = \lim_{x \rightarrow 2} f(x)$.

$$\begin{aligned} \lim_{x \rightarrow 2} f(x) &= \lim_{x \rightarrow 2} \frac{6x^2 - 13x + 2}{2x^2 - 8} = \lim_{x \rightarrow 2} \frac{(6x-1)(x-2)}{2(x+2)(x-2)} \\ &= \lim_{x \rightarrow 2} \frac{6x-1}{2(x+2)} = \frac{11}{8} \end{aligned}$$

$f(2)$ should be defined as $\frac{11}{8}$ to make f continuous at $x = 2$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response .

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0	1
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The student response accurately indicates $x = -2$ with reason.

Solution:

The graph of f has a vertical asymptote at $x = -2$ because the denominator $2x^2 - 8$ equals 0 at $x = -2$ but the numerator does not equal 0 at $x = -2$.

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3. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$f(x) = \begin{cases} \frac{2x^2+5x-3}{x^2+4x+3} & \text{for } x < -3 \\ kx + \frac{1}{2} & \text{for } -3 \leq x \leq 0 \\ \frac{2^x}{3^{x-1}} & \text{for } x > 0 \end{cases}$$

Let f be the function defined above, where k is a constant.

- For what value of k , if any, is f continuous at $x = -3$? Justify your answer.
- What type of discontinuity does f have at $x = 0$? Give a reason for your answer.
- Find all horizontal asymptotes to the graph of f . Show the work that leads to your answer.

Part A

At most 2 out of 3 points if mathematical notation for limits is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ $\lim_{x \rightarrow -3^-} f(x)$
- ☐ $\lim_{x \rightarrow -3^+} f(x)$
- ☐ justification

Solution:

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$$\begin{aligned}\lim_{x \rightarrow -3^-} f(x) &= \lim_{x \rightarrow -3^-} \frac{2x^2 + 5x - 3}{x^2 + 4x + 3} = \lim_{x \rightarrow -3^-} \frac{(2x-1)(x+3)}{(x+1)(x+3)} \\ &= \lim_{x \rightarrow -3^-} \frac{2x-1}{x+1} = \frac{\lim_{x \rightarrow -3^-} (2x-1)}{\lim_{x \rightarrow -3^-} (x+1)} \\ &= \frac{2(-3)-1}{-3+1} = \frac{7}{2}\end{aligned}$$

$$\lim_{x \rightarrow -3^+} f(x) = \lim_{x \rightarrow -3^+} \left(kx + \frac{1}{2}\right) = -3k + \frac{1}{2}$$

$$f(-3) = -3k + \frac{1}{2}$$

$$f \text{ is continuous at } x = -3 \text{ if } \lim_{x \rightarrow -3^-} f(x) = \lim_{x \rightarrow -3^+} f(x) = f(-3).$$

$$-3k + \frac{1}{2} = \frac{7}{2} \Rightarrow -3k = 3 \Rightarrow k = -1$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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he student response accurately includes both of the criteria below.

- ☐ $\lim_{x \rightarrow 0^+} f(x)$
- ☐ answer

Solution:

$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{2^x}{3^{x-1}} = \infty$ because the numerator approaches 1, while the denominator approaches 0 from the right.

Therefore, f has a discontinuity due to a vertical asymptote at $x = 0$.

Part C

At most 1 out of 2 points if both correct asymptotes are listed without reference to limits OR if one correct asymptote is listed without any incorrect asymptotes and/or reference to limits.

Students do not have to show the factorization to earn the second point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

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0	1	2
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The student response accurately includes both of the criteria below.

- ☐ $\lim_{x \rightarrow \infty} f(x)$
- ☐ $\lim_{x \rightarrow -\infty} f(x)$

Solution:

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{2^x}{3^x - 1} = \lim_{x \rightarrow \infty} \frac{2^x}{3^x} = \lim_{x \rightarrow \infty} \left(\frac{2}{3}\right)^x = 0 \text{ because } 0 < \frac{2}{3} < 1.$$

$$\begin{aligned} \lim_{x \rightarrow -\infty} f(x) &= \lim_{x \rightarrow -\infty} \frac{2x^2 + 5x - 3}{x^2 + 4x + 3} = \lim_{x \rightarrow -\infty} \frac{x^2 \left(2 + \frac{5}{x} - \frac{3}{x^2}\right)}{x^2 \left(1 + \frac{4}{x} + \frac{3}{x^2}\right)} \\ &= \lim_{x \rightarrow -\infty} \frac{2 + \frac{5}{x} - \frac{3}{x^2}}{1 + \frac{4}{x} + \frac{3}{x^2}} = \frac{\lim_{x \rightarrow -\infty} \left(2 + \frac{5}{x} - \frac{3}{x^2}\right)}{\lim_{x \rightarrow -\infty} \left(1 + \frac{4}{x} + \frac{3}{x^2}\right)} \\ &= \frac{2 + 0 - 0}{1 + 0 + 0} = 2 \end{aligned}$$

The graph of f has horizontal asymptotes at $y = 0$ and $y = 2$.

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4. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$f(x) = \begin{cases} \frac{x^2+kx-2}{3x^2+4x+3} & \text{for } x < -2 \\ x^3 + 2 & \text{for } -2 \leq x < 0 \\ 0 & \text{for } x = 0 \\ \frac{2e^x}{2-e^x} & \text{for } x > 0 \end{cases}$$

Let f be the function defined above, where k is a constant.

- For what value of k , if any, is f continuous at $x = -2$? Justify your answer.
- What type of discontinuity does f have at $x = 0$? Give a reason for your answer.
- Find all horizontal asymptotes to the graph of f . Show the work that leads to your answer.

Part A

At most 2 out of 3 points if mathematical notation for limits is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ $\lim_{x \rightarrow -2^-} f(x)$
- ☐ $\lim_{x \rightarrow -2^+} f(x)$
- ☐ justification equating limit with $f(-2)$

Solution:

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$$\begin{aligned}\lim_{x \rightarrow -2^-} f(x) &= \lim_{x \rightarrow -2^-} \frac{x^2 + kx - 2}{3x^2 + 4x + 3} = \frac{\lim_{x \rightarrow -2^-} (x^2 + kx - 2)}{\lim_{x \rightarrow -2^-} (3x^2 + 4x + 3)} \\ &= \frac{(-2)^2 + k(-2) - 2}{3(-2)^2 + 4(-2) + 3} = \frac{2 - 2k}{7}\end{aligned}$$

$$\lim_{x \rightarrow -2^+} f(x) = \lim_{x \rightarrow -2^+} (x^3 + 2) = (-2)^3 + 2 = -6$$

$$f(-2) = -6$$

$$f \text{ is continuous at } x = -2 \text{ if } \lim_{x \rightarrow -2^-} f(x) = \lim_{x \rightarrow -2^+} f(x) = f(-2).$$

$$\frac{2 - 2k}{7} = -6 \Rightarrow 2 - 2k = -42 \Rightarrow k = 22$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

☐ $\lim_{x \rightarrow 0^-} f(x) \text{ and } \lim_{x \rightarrow 0^+} f(x)$

☐ answer
Solution:

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (x^3 + 2) = 2$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{2e^x}{2 - e^x} = \frac{2e^0}{2 - e^0} = 2$$

Because $\lim_{x \rightarrow 0} f(x) = 2$ and $f(0) = 0$, f has a removable discontinuity at $x = 0$.

Part C

At most 1 out of 2 points if both correct asymptotes are listed without reference to limits OR if one correct asymptote is listed without any incorrect asymptotes and/or reference to limits.

Students do not have to show the factorization to earn the second point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

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0	1	2
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The student response accurately includes both of the criteria below.

- ☐ $\lim_{x \rightarrow \infty} f(x)$
- ☐ $\lim_{x \rightarrow -\infty} f(x)$

Solution:

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{2e^x}{2 - e^x} = \lim_{x \rightarrow \infty} \frac{2e^x}{-e^x} = \lim_{x \rightarrow \infty} (-2) = -2$$

$$\begin{aligned} \lim_{x \rightarrow -\infty} f(x) &= \lim_{x \rightarrow -\infty} \frac{x^2 + kx - 2}{3x^2 + 4x + 3} = \lim_{x \rightarrow -\infty} \frac{x^2 \left(1 + \frac{k}{x} - \frac{2}{x^2}\right)}{x^2 \left(3 + \frac{4}{x} + \frac{3}{x^2}\right)} \\ &= \lim_{x \rightarrow -\infty} \frac{1 + \frac{k}{x} - \frac{2}{x^2}}{3 + \frac{4}{x} + \frac{3}{x^2}} = \frac{\lim_{x \rightarrow -\infty} \left(1 + \frac{k}{x} - \frac{2}{x^2}\right)}{\lim_{x \rightarrow -\infty} \left(3 + \frac{4}{x} + \frac{3}{x^2}\right)} \\ &= \frac{1 + 0 - 0}{3 + 0 + 0} = \frac{1}{3} \end{aligned}$$

The graph of f has horizontal asymptotes at $y = -2$ and $y = \frac{1}{3}$.

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5. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be the function defined by $f(x) = \frac{cx-5x^2}{2x^2+ax+b}$, where a , b , and c are constants. The graph of f has a vertical asymptote at $x = 1$, and f has a removable discontinuity at $x = -2$.

- Show that $a = 2$ and $b = -4$.
- Find the value of c . Justify your answer.
- To make f continuous at $x = -2$, $f(-2)$ should be defined as what value? Justify your answer.
- Write an equation for the horizontal asymptote to the graph of f . Show the work that leads to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ sets $2x^2 + ax + b = 0$ for $x = -2$ and $x = 1$
- ☐ verification

Solution:

For f to have a removable discontinuity at $x = -2$ the denominator must equal zero at $x = -2$.

$$2(-2)^2 + a(-2) + b = 0 \Rightarrow 2a - b = 8$$

For f to have a vertical asymptote at $x = 1$ the denominator must equal zero at $x = 1$.

$$2(1)^2 + a(1) + b = 0 \Rightarrow a + b = -2$$

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$$2a - b = 8 \text{ and } a + b = -2 \Rightarrow 3a = 6 \Rightarrow a = 2$$

$$4 - b = 8 \Rightarrow b = 4 - 8 = -4$$

Part B

Work for the justification point may come before the answer.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ answer
- ☐ justification

Solution:

Since f has a removable discontinuity at $x = -2$ and the denominator equals 0 at $x = -2$, the numerator must also equal 0 at $x = -2$.

$$0 = c(-2) - 5(-2)^2 = -2c - 20$$

$$\Rightarrow c = \frac{20}{-2} = -10$$

Part C

At most 1 out of 2 points if mathematical notation for any limit is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ uses $\lim_{x \rightarrow -2} f(x)$
- ☐ answer with justification

Solution:

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f will be continuous at $x = -2$ if $f(-2) = \lim_{x \rightarrow -2} f(x)$.

$$\begin{aligned}\lim_{x \rightarrow -2} f(x) &= \lim_{x \rightarrow -2} \frac{-10x - 5x^2}{2x^2 + 2x - 4} = \lim_{x \rightarrow -2} \frac{-5x(2+x)}{2(x-1)(x+2)} \\ &= \lim_{x \rightarrow -2} \frac{-5x}{2(x-1)} = \frac{-5(-2)}{2(-2-1)} = \frac{10}{-6} = -\frac{5}{3}\end{aligned}$$

$f(-2)$ should be defined as $-\frac{5}{3}$ to make f continuous at $x = -2$.

Part D

No point if correct asymptote is listed without reference to limits.

Students do not have to show the factorization to earn the point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
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The student response accurately includes an equation for the horizontal asymptote.

Solution:

$$\begin{aligned}\lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} \frac{-10x - 5x^2}{2x^2 + 2x - 4} = \lim_{x \rightarrow \infty} \frac{x^2(-\frac{10}{x} - 5)}{x^2(2 + \frac{2}{x} - \frac{4}{x^2})} \\ &= \lim_{x \rightarrow \infty} \frac{-\frac{10}{x} - 5}{2 + \frac{2}{x} - \frac{4}{x^2}} = \frac{\lim_{x \rightarrow \infty} (-\frac{10}{x} - 5)}{\lim_{x \rightarrow \infty} (2 + \frac{2}{x} - \frac{4}{x^2})} \\ &= \frac{0 - 5}{2 + 0 - 0} = -\frac{5}{2}\end{aligned}$$

The horizontal asymptote to the graph of f is $y = -\frac{5}{2}$.

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6. Let f be a function defined by $f(x) = \begin{cases} 1 - 2 \sin x & \text{for } x \leq 0 \\ e^{-4x} & \text{for } x > 0. \end{cases}$

(a) Show that f is continuous at $x = 0$.

(b) For $x \neq 0$, express $f'(x)$ as a piecewise-defined function. Find the value of x for which $f'(x) = -3$.

(c) Find the average value of f on the interval $[-1, 1]$.

Part A

2 point is earned for analysis

$$\lim_{x \rightarrow 0^-} (1 - 2 \sin x) = 1$$

$$\lim_{x \rightarrow 0^+} e^{-4x} = 1$$

$$f(0) = 1$$

$$\text{So, } \lim_{x \rightarrow 0} f(x) = f(0).$$

Therefore f is continuous at $x = 0$.



0	1	2
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The student response earns all of the following points:

2 point is earned for analysis

$$\lim_{x \rightarrow 0^-} (1 - 2 \sin x) = 1$$

$$\lim_{x \rightarrow 0^+} e^{-4x} = 1$$

$$f(0) = 1$$

$$\text{So, } \lim_{x \rightarrow 0} f(x) = f(0).$$

Therefore f is continuous at $x = 0$.

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Part B

2 point is earned for $f'(x)$

1 point is earned for value of x

$$f'(x) = \begin{cases} -2 \cos x & \text{for } x < 0 \\ -4e^{-4x} & \text{for } x > 0. \end{cases}$$

$-2 \cos x \neq -3$ for all values for $x < 0$.

$$-4e^{-4x} = -3 \text{ when } x = -\frac{1}{4} \ln\left(\frac{3}{4}\right) > 0.$$

$$\text{Therefore, } f'(x) = -3 \text{ for } x = -\frac{1}{4} \ln\left(\frac{3}{4}\right).$$



0	1	2	3
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The student response earns all of the following points:

2 point is earned for $f'(x)$

1 point is earned for value of x

$$f'(x) = \begin{cases} -2 \cos x & \text{for } x < 0 \\ -4e^{-4x} & \text{for } x > 0. \end{cases}$$

$-2 \cos x \neq -3$ for all values for $x < 0$.

$$-4e^{-4x} = -3 \text{ when } x = -\frac{1}{4} \ln\left(\frac{3}{4}\right) > 0.$$

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Therefore, $f'(x) = -3$ for $x = -\frac{1}{4}\ln\left(\frac{3}{4}\right)$.

Part C

1 point is earned for $\int_{-1}^0 (1 - 2 \sin x) dx$ and $\int_0^1 e^{-4x} dx$

2 point is earned for antiderivatives

1 point is earned for answer

$$\begin{aligned} \int_{-1}^1 f(x) dx &= \int_{-1}^0 f(x) dx + \int_0^1 f(x) dx \\ &= \int_{-1}^0 (1 - 2 \sin x) dx + \int_0^1 e^{-4x} dx \\ &= [x + 2 \cos]_{x=-1}^{x=0} + \left[-\frac{1}{4}e^{-4x}\right]_{x=0}^{x=1} \\ &= (3 - 2 \cos(-1)) + \left(-\frac{1}{4}e^{-4} + \frac{1}{4}\right) \end{aligned}$$

$$\begin{aligned} \text{Average value} &= \frac{1}{2} \int_{-1}^1 f(x) dx \\ &= \frac{13}{8} - \cos(-1) - \frac{1}{8}e^{-4} \end{aligned}$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for $\int_{-1}^0 (1 - 2 \sin x) dx$ and $\int_0^1 e^{-4x} dx$

2 point is earned for antiderivatives

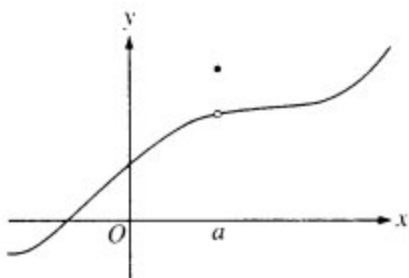
1 point is earned for answer

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$$\begin{aligned}
 \int_{-1}^1 f(x) dx &= \int_{-1}^0 f(x) dx + \int_0^1 f(x) dx \\
 &= \int_{-1}^0 (1 - 2 \sin x) dx + \int_0^1 e^{-4x} dx \\
 &= [x + 2 \cos x]_{x=-1}^{x=0} + \left[-\frac{1}{4}e^{-4x}\right]_{x=0}^{x=1} \\
 &= (3 - 2 \cos(-1)) + \left(-\frac{1}{4}e^{-4} + \frac{1}{4}\right)
 \end{aligned}$$

$$\begin{aligned}
 \text{Average value} &= \frac{1}{2} \int_{-1}^1 f(x) dx \\
 &= \frac{13}{8} - \cos(-1) - \frac{1}{8}e^{-4}
 \end{aligned}$$

7.



The graph of a function f is shown above. Which of the following statements about f is false?

- (A) f is continuous at $x = a$. ✓
- (B) f has a relative maximum at $x = a$.
- (C) $x = a$ is in the domain of f .
- (D) $\lim_{x \rightarrow a^+} f(x)$ is equal to $\lim_{x \rightarrow a^-} f(x)$
- (E) $\lim_{x \rightarrow a} f(x)$ exists

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8. Let
- f
- be the function defined by

$$f(x) = \begin{cases} \sqrt{x+1} & \text{for } 0 \leq x \leq 3 \\ 5-x & \text{for } 3 < x \leq 5. \end{cases}$$

- (a) Is f continuous at $x = 3$? Explain why or why not.
- (b) Find the average value of $f(x)$ on the closed interval $0 \leq x \leq 5$.
- (c) Suppose the function g is defined by

$$g(x) = \begin{cases} k\sqrt{x+1} & \text{for } 0 \leq x \leq 3 \\ mx+2 & \text{for } 3 < x \leq 5, \end{cases}$$

where k and m are constants. If g is differentiable at $x = 3$, what are the values of k and m ?

Part A

1 point is earned for answering "yes" and equating the values of the left- and right-hand limits

1 point is earned for the correct explanation involving limits

f is continuous at $x = 3$ because $\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x) = 2$.

Therefore, $\lim_{x \rightarrow 3} f(x) = 2 = f(3)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for answering "yes" and equating the values of the left- and right-hand limits

1 point is earned for the correct explanation involving limits

f is continuous at $x = 3$ because $\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x) = 2$.

Therefore, $\lim_{x \rightarrow 3} f(x) = 2 = f(3)$.

Part B

1 point is earned for $k \int_0^3 f(x)dx + k \int_3^5 f(x)dx$ (where $k \neq 0$)

$$\int_0^5 f(x)dx = \int_0^3 f(x)dx + \int_3^5 f(x)dx$$

1 point is earned for the correct antiderivative of $\sqrt{x+1}$

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1 point is earned for the correct antiderivative of $5 - x$

$$= \frac{2}{3}(x+1)^{3/2} \Big|_0^3 + \left(5x - \frac{1}{2}x^2\right) \Big|_3^5$$

$$= \left(\frac{16}{3} - \frac{2}{3}\right) + \left(\frac{25}{2} - \frac{21}{2}\right) = \frac{20}{3}$$

1 point is earned for the correct evaluation and answer

Average value: $\frac{1}{5} \int_0^5 f(x) dx = \frac{4}{3}$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for $k \int_0^3 f(x) dx + k \int_3^5 f(x) dx$ (where $k \neq 0$)

$$\int_0^5 f(x) dx = \int_0^3 f(x) dx + \int_3^5 f(x) dx$$

1 point is earned for the correct antiderivative of $\sqrt{x+1}$

1 point is earned for the correct antiderivative of $5 - x$

$$= \frac{2}{3}(x+1)^{3/2} \Big|_0^3 + \left(5x - \frac{1}{2}x^2\right) \Big|_3^5$$

$$= \left(\frac{16}{3} - \frac{2}{3}\right) + \left(\frac{25}{2} - \frac{21}{2}\right) = \frac{20}{3}$$

1 point is earned for the correct evaluation and answer

Average value: $\frac{1}{5} \int_0^5 f(x) dx = \frac{4}{3}$

Part C

1 point is earned for the correct answer $2k = 3m + 2$

Since g is continuous at $x = 3$, $2k = 3m + 2$.

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$$g'(x) = \begin{cases} \frac{k}{2\sqrt{x+1}} & \text{for } 0 < x < 3 \\ m & \text{for } 3 < x < 5 \end{cases}$$

$$\lim_{x \rightarrow 3^-} g'(x) = \frac{k}{4} \text{ and } \lim_{x \rightarrow 3^+} g'(x) = m$$

1 point is earned for the correct answer $\frac{k}{4} = m$

Since these two limits exist and g is differentiable at $x = 3$, the two limits are equal. Thus $\frac{k}{4} = m$.

1 point is earned for the correct values for k and m

$$8m = 3m + 2; m = \frac{2}{5} \text{ and } k = \frac{8}{5}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct answer $2k = 3m + 2$

Since g is continuous at $x = 3$, $2k = 3m + 2$.

$$g'(x) = \begin{cases} \frac{k}{2\sqrt{x+1}} & \text{for } 0 < x < 3 \\ m & \text{for } 3 < x < 5 \end{cases}$$

$$\lim_{x \rightarrow 3^-} g'(x) = \frac{k}{4} \text{ and } \lim_{x \rightarrow 3^+} g'(x) = m$$

1 point is earned for the correct answer $\frac{k}{4} = m$

Since these two limits exist and g is differentiable at $x = 3$, the two limits are equal. Thus $\frac{k}{4} = m$.

1 point is earned for the correct values for k and m

$$8m = 3m + 2; m = \frac{2}{5} \text{ and } k = \frac{8}{5}$$

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9. Let f be the function defined by $f(x) = \begin{cases} \sqrt{x+1} & \text{for } 0 \leq x \leq 3 \\ 5-x & \text{for } 3 < x \leq 5. \end{cases}$

Is f continuous at $x = 3$? Explain why or why not.

Part A

1 point is earned for answering "yes" and equating the values of the left- and right-hand limits

1 point is earned for the correct explanation involving limits

f is continuous at $x = 3$ because $\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x) = 2$.

Therefore, $\lim_{x \rightarrow 3} f(x) = 2 = f(3)$.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for answering "yes" and equating the values of the left- and right-hand limits

1 point is earned for the correct explanation involving limits

f is continuous at $x = 3$ because $\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x) = 2$.

Therefore, $\lim_{x \rightarrow 3} f(x) = 2 = f(3)$.

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10. Let f be a function defined by $f(x) = \begin{cases} 2x - x^2 & \text{for } x \leq 1, \\ x^2 + kx + p & \text{for } x > 1. \end{cases}$
- (a) For what values of k and p will f be continuous and differentiable at $x = 1$?
- (b) For the values of k and p found in part (a), on what interval or intervals is f increasing?
- (c) Using the values of k and p found in part (a), find all points of inflection of the graph of f . Support your conclusion.

Part A

1 point is earned for equation of $f(1)$

1 point is earned for $1 = 1 + k + p$

1 point is earned for $0 = 2 + k$

1 point is earned for solution of system

For continuity at $x = 1$,

$$\lim_{x \rightarrow 1^-} (2x - x^2) = f(1) = \lim_{x \rightarrow 1^+} (x^2 + kx + p)$$

Therefore $1 = 1 + k + p$

Since f is continuous at $x = 1$ and is piecewise polynomial, left and right derivatives exist.

$$f'_-(1) = 0 \text{ and } f'_+(1) = 2 + k$$

For differentiability at $x = 1$, $0 = 2 + k$.

Therefore $k = -2$, $p = 2$.



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for equation of $f(1)$

1 point is earned for $1 = 1 + k + p$

1 point is earned for $0 = 2 + k$

1 point is earned for solution of system

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For continuity at $x = 1$,

$$\lim_{x \rightarrow 1^-} (2x - x^2) = f(1) = \lim_{x \rightarrow 1^+} (x^2 + kx + p)$$

Therefore $1 = 1 + k + p$

Since f is continuous at $x = 1$ and is piecewise polynomial, left and right derivatives exist.

$$f'_-(1) = 0 \text{ and } f'_+(1) = 2 + k$$

For differentiability at $x = 1$, $0 = 2 + k$.

Therefore $k = -2$, $p = 2$.

Part B

1 point is earned for correct f' and analysis of sign of f' on domain

1 point is earned for $(-\infty, \infty)$

$$f'(x) = 2 - 2x, x \leq 1$$

$$2 - 2x > 0$$

$$x < 1$$

$$f'(x) = 2x - 2, x > 1$$

$$2x - 2 > 0$$

$$x > 1$$

Since f increases on each of $(-\infty, 1)$ and $(1, \infty)$ and is continuous at $x = 1$, f is increasing on $(-\infty, \infty)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correct f' and analysis of sign of f' on domain

1 point is earned for $(-\infty, \infty)$

$$f'(x) = 2 - 2x, x \leq 1$$

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$$2 - 2x > 0$$

$$x < 1$$

$$f'(x) = 2x - 2, x > 1$$

$$2x - 2 > 0$$

$$x > 1$$

Since f increases on each of $(-\infty, 1)$ and $(1, \infty)$ and is continuous at $x = 1$, f is increasing on $(-\infty, \infty)$.

Part C

1 point is earned for f''

1 point is earned for noting sign changes in f'' or change of concavity in f

1 point is earned for ordered pair (or equivalent)

$$f''(x) = -2, x < 1$$

$$f''(x) = 2, x > 1$$

Since $f''(x) < 0$ on $(-\infty, 1)$ and

$f''(x) > 0$ on $(1, \infty)$ and

$f(1)$ is defined,

$(1, f(1)) = (1, 1)$ is a point of inflection.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for f''

1 point is earned for noting sign changes in f'' or change of concavity in f

1 point is earned for ordered pair (or equivalent)

$$f''(x) = -2, x < 1$$

$$f''(x) = 2, x > 1$$

Since $f''(x) < 0$ on $(-\infty, 1)$ and

Unit 1 AP Topics 9-12

$f''(x) > 0$ on $(1, \infty)$ and

$f(1)$ is defined,

$(1, f(1)) = (1, 1)$ is a point of inflection.

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11. Let f be a function defined by $f(x) = \begin{cases} 2x - x^2 & \text{for } x \leq 1 \\ x^2 + kx + p & \text{for } x > 1 \end{cases}$.

For what values of k and p will f be continuous and differentiable at $x=1$?

Part A

1 point is earned for equation of $f(1)$

1 point is earned for $1 = 1 + k + p$

1 point is earned for $0 = 2 + k$

1 point is earned for solution of system

For continuity at $x = 1$,

$$\lim_{x \rightarrow 1^-} (2x - x^2) = f(1) = \lim_{x \rightarrow 1^+} (x^2 + kx + p)$$

Therefore $1 = 1 + k + p$

Since f is continuous at $x=1$ and is piecewise polynomial, left and right derivatives exist.

$$f'_-(1) = 0 \text{ and } f'_+(1) = 2 + k.$$

For differentiability at $x = 1$, $0 = 2 + k$.

Therefore $k = -2, p = 2$



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for equation of $f(1)$

1 point is earned for $1 = 1 + k + p$

1 point is earned for $0 = 2 + k$

1 point is earned for solution of system

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For continuity at $x = 1$,

$$\lim_{x \rightarrow 1^-} (2x - x^2) = f(1) = \lim_{x \rightarrow 1^+} (x^2 + kx + p)$$

$$\text{Therefore } 1 = 1 + k + p$$

Since f is continuous at $x=1$ and is piecewise polynomial, left and right derivatives exist.

$$f_-^1(1) = 0 \text{ and } f_+^1(1) = 2 + k.$$

For differentiability at $x = 1$, $0 = 2 + k$.

$$\text{Therefore } k = -2, p = 2$$

12. At $x = 3$, the function given by $f(x) = \begin{cases} x^2, & x < 3 \\ 6x - 9, & x \geq 3 \end{cases}$ is

- (A) undefined
- (B) continuous but not differentiable
- (C) differentiable but not continuous
- (D) neither continuous nor differentiable
- (E) both continuous and differentiable



13. If $\lim_{x \rightarrow 3} f(x) = 7$, which of the following must be true?

- I. f is continuous at $x = 3$
- II. f is differentiable at $x = 3$
- III. $f(3) = 7$

- (A) None
- (B) II only
- (C) III only
- (D) I and III only
- (E) I, II, and III



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14. If $\lim_{x \rightarrow a} f(x) = L$, where L is a real number, which of the following must be true?

- (A) $f(a)$ exists
- (B) $f(x)$ is continuous at $x=a$.
- (C) $f(x)$ is defined at $x=a$.
- (D) $f(a)=L$

(E) none of the above



15.  Which of the following functions are continuous at $x=1$?

I. $\ln x$

II. e^x

III. $\ln(e^x-1)$

- (A) I only
- (B) II only
- (C) I and II only
- (D) II and III only

(E) I, II, and III



16.

I. If f is defined at $x = a$, then $\lim_{x \rightarrow a} f(x) = f(a)$.

II. If f is continuous at $x = a$, then $\lim_{x \rightarrow a} f(x) = f(a)$.

III. If f is differentiable at $x = a$, then $\lim_{x \rightarrow a} f(x) = f(a)$.

For any function f , which of the following statements must be true?

- (A) III only
- (B) I and II only
- (C) II and III only
- (D) I, II, and III



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17.

a	$\lim_{x \rightarrow a^-} f(x)$	$\lim_{x \rightarrow a^+} f(x)$	$f(a)$
-1	4	6	4
0	-3	-3	5
1	2	2	2

The function f has the properties indicated in the table above. Which of the following must be true?

- (A) f is continuous at $x = -1$.
(B) f is continuous at $x = 0$.
(C) f is continuous at $x = 1$.
(D) f is differentiable at $x = 0$.
(E) f is differentiable at $x = 1$.



18. $f(x) = \begin{cases} x^2 - 3x + 9 & \text{for } x \leq 2 \\ kx + 1 & \text{for } x > 2 \end{cases}$

The function f is defined above. For what value of k , if any, is f continuous at $x = 2$?

- (A) 1
(B) 2
(C) 3
(D) 7
(E) No value of k will make f continuous at $x = 2$.



19. If $\lim_{x \rightarrow a} f(x) = f(a)$, then which of the following statements about f must be true?

- (A) f is continuous at $x = a$.
(B) f is differentiable at $x = a$.
(C) For all values of x , $f(x) = f(a)$.
(D) The line $y = f(a)$ is tangent to the graph of f at $x = a$.
(E) The line $x = a$ is a vertical asymptote of the graph of f .



20. If the function f is continuous for all real numbers and if $f(x) = \frac{x^2 - 4}{x + 2}$ when $x \neq -2$, then $f(-2) =$

- (A) -4
(B) -2
(C) -1
(D) 0
(E) 2



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21. The function f is defined by $f(x) = \sqrt{25 - x^2}$ for $-5 \leq x \leq 5$.

(a) Find $f'(x)$.

(b) Write an equation for the line tangent to the graph of f at $x = -3$.

(c) Let g be the function defined by $g(x) = \begin{cases} f(x) & \text{for } -5 \leq x \leq -3 \\ x + 7 & \text{for } -3 < x \leq 5. \end{cases}$

Is g continuous at $x = -3$? Use the definition of continuity to explain your answer.

(d) Find the value of $\int_0^5 x\sqrt{25 - x^2} dx$.

Part A

2 points are earned for $f'(x)$

$$f'(x) = \frac{1}{2}(25 - x^2)^{-1/2}(-2x) = \frac{-x}{\sqrt{25 - x^2}}, \quad -5 < x < 5$$



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for $f'(x)$

$$f'(x) = \frac{1}{2}(25 - x^2)^{-1/2}(-2x) = \frac{-x}{\sqrt{25 - x^2}}, \quad -5 < x < 5$$

Part B

1 point is earned for $f'(-3)$

1 point is earned for the answer

$$f'(-3) = \frac{3}{\sqrt{25-9}} = \frac{3}{4}$$

$$f(-3) = \sqrt{25 - 9} = 4$$

An equation for the tangent line is $y = 4 + \frac{3}{4}(x + 3)$.

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0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f'(-3)$

1 point is earned for the answer

$$f'(-3) = \frac{3}{\sqrt{25-9}} = \frac{3}{4}$$

$$f(-3) = \sqrt{25-9} = 4$$

An equation for the tangent line is $y = 4 + \frac{3}{4}(x + 3)$.

Part C

1 point is earned for considering one-sided limits

1 point is earned for the answer with explanation

$$\lim_{x \rightarrow -3^-} g(x) = \lim_{x \rightarrow -3^-} f(x) = \lim_{x \rightarrow -3^-} \sqrt{25 - x^2} = 4$$

$$\lim_{x \rightarrow -3^+} g(x) = \lim_{x \rightarrow -3^+} (x + 7) = 4$$

Therefore, $\lim_{x \rightarrow -3} g(x) = 4$.

$$g(-3) = f(-3) = 4$$

So, $\lim_{x \rightarrow -3} g(x) = g(-3)$.

Therefore, g is continuous at $x = -3$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for considering one-sided limits

1 point is earned for the answer with explanation

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$$\lim_{x \rightarrow -3^-} g(x) = \lim_{x \rightarrow -3^-} f(x) = \lim_{x \rightarrow -3^-} \sqrt{25 - x^2} = 4$$

$$\lim_{x \rightarrow -3^+} g(x) = \lim_{x \rightarrow -3^+} (x + 7) = 4$$

Therefore, $\lim_{x \rightarrow -3} g(x) = 4$.

$$g(-3) = f(-3) = 4$$

So, $\lim_{x \rightarrow -3} g(x) = g(-3)$.

Therefore, g is continuous at $x = -3$.

Part D

2 points are earned for antiderivative

1 point is earned for the answer

$$\text{Let } u = 25 - x^2 \Rightarrow du = -2x dx$$

$$\begin{aligned} \int_0^5 x \sqrt{25 - x^2} dx &= -\frac{1}{2} \int_{25}^0 \sqrt{u} du \\ &= \left[-\frac{1}{2} \cdot \frac{2}{3} u^{3/2} \right]_{u=25}^{u=0} \\ &= -\frac{1}{3} (0 - 125) = \frac{125}{3} \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for antiderivative

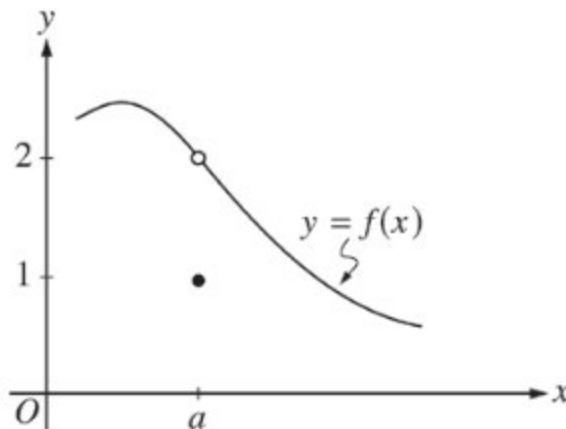
1 point is earned for the answer

$$\text{Let } u = 25 - x^2 \Rightarrow du = -2x dx$$

$$\begin{aligned} \int_0^5 x \sqrt{25 - x^2} dx &= -\frac{1}{2} \int_{25}^0 \sqrt{u} du \\ &= \left[-\frac{1}{2} \cdot \frac{2}{3} u^{3/2} \right]_{u=25}^{u=0} \\ &= -\frac{1}{3} (0 - 125) = \frac{125}{3} \end{aligned}$$

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22.



The graph of a function f is shown in the figure above. Which of the following statements is true?

- (A) $f(a) = 2$
 (B) f is continuous at $x = a$.
 (C) $\lim_{x \rightarrow a} f(x) = 1$
 (D) $\lim_{x \rightarrow a} f(x) = 2$
 (E) $\lim_{x \rightarrow a} f(x)$ does not exist.



23.

Let f be the function defined by $f(x) = \begin{cases} x^2 + 2 & \text{for } x \leq 3, \\ 6x + k & \text{for } x > 3. \end{cases}$

If f is continuous at $x = 3$, what is the value of k ?

- (A) -7
 (B) 2
 (C) 3
 (D) 7
 (E) There is no such value of k .



24.

If f is a continuous function on $[a, b]$, which of the following is necessarily true?

- (A) f' exists on (a, b)
 (B) If $f(x_0)$ is a maximum of f , then $f'(x_0) = 0$.
 (C) $\lim_{x \rightarrow x_0} f(x) = f\left(\lim_{x \rightarrow x_0} x\right)$ for $x_0 \in (a, b)$
 (D) $f'(x) = 0$ for some $x \in [a, b]$
 (E) The graph of f' is a straight line.



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25. Let f be the function given by $f(x) = \frac{(x-2)^2(x+3)}{(x-2)(x+1)}$. For which of the following values of x is f not continuous?

- (A) -3 and -1 only
- (B) -3, -1, and 2
- (C) -1 only
- (D) -1 and 2 only
- (E) 2 only



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26. Let f be the function given by $f(x) = \ln(x / x - 1)$
What is the domain of f ?

Part A

1 point is earned for setting $\frac{x}{x-1} > 0$

1 point is earned for first interior point

1 point is earned for second interior point

1 point is **deducted** for each incorrect point

$$\frac{x}{x-1} > 0$$

$$x > 0 \text{ and } x - 1 > 0 \Rightarrow x > 1$$

$$x < 0 \text{ and } x - 1 < 0 \Rightarrow x < 0$$

$$x < 0 \text{ or } x > 1$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for setting $\frac{x}{x-1} > 0$

1 point is earned for first interior point

1 point is earned for second interior point

1 point is **deducted** for each incorrect point

$$\frac{x}{x-1} > 0$$

$$x > 0 \text{ and } x - 1 > 0 \Rightarrow x > 1$$

$$x < 0 \text{ and } x - 1 < 0 \Rightarrow x < 0$$

$$x < 0 \text{ or } x > 1$$

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27. What is the domain of the function f given by $f(x) = \frac{\sqrt{x^2-4}}{x-3}$?

(A) $\{x : x \neq 3\}$

(B) $\{x : |x| \leq 2\}$

(C) $\{x : |x| \geq 2\}$

(D) $\{x : |x| \geq 2 \text{ and } x \neq 3\}$



(E) $\{x : x \geq 2 \text{ and } x \neq 3\}$

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28. A tank contains 30 liters of water at time $t = 0$. For time $0 \leq t \leq 20$ minutes, water flows into the tank at a rate modeled by the function f defined by $f(t) = 2 \ln(t + 3) - 1.6$. For time $20 < t \leq 45$ minutes, water flows into the tank at a rate modeled by the function g defined by $g(t) = \frac{7.5}{\ln t} + 2.2$. Both $f(t)$ and $g(t)$ are measured in liters per minute.

**Part A**

Find the rate of change of g at time $t = 23$ minutes. Show the setup for your calculations, and indicate units of measure.

Part B

How many liters of water are in the tank at time $t = 20$ minutes? Show the setup for your calculations.

Part C

Let w be the function defined by $w(t) = \begin{cases} f(t) & \text{for } 0 \leq t \leq 20 \\ g(t) & \text{for } 20 < t \leq 45 \end{cases}$. Find $\lim_{t \rightarrow 20^+} w(t)$. Is w continuous at $t = 20$? Use the definition of continuity to justify your answer.

Part D

Consider the function w defined in part C. Find the average value of w from time $t = 15$ minutes to time $t = 30$ minutes. Show the setup for your calculations.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P1** is earned for a correct value either rounded or truncated to three places after the decimal and for labeling the value as $g'(23)$. An inappropriately rounded answer does not earn the point.
- **P1** can also be earned for the exact value $\frac{-7.5}{23(\ln 23)^2}$.

Model Solution:

$$g'(23) = -0.033$$



0	1
---	---

The response accurately includes the criteria below.

- $g'(23)$

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Part A Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P2** is earned for correct units, whether or not they are attached to a numerical value for the rate of change.
- **P2** is also earned for the units “liters/minute².”

Model Solution:

The rate of change in the rate at time $t = 23$ is -0.033 liters per minute per minute.



0	1
---	---

The response accurately includes the criteria below.

- Units

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P3** is earned with a definite integral containing the correct integrand and correct lower and upper limits, with or without the differential.
- If the differential dt is missing:
 - Writing $\int_0^{20} f(t)$ or $30 + \int_0^{20} f(t)$ earns **P3**.
 - Because $\int_0^{20} f(t) + 30$ may mean $\int_0^{20} f(t)dt + 30$ or $\int_0^{20} (f(t) + 30)dt$:
 - Writing $\int_0^{20} f(t) + 30$ in the presence of a correct answer of 95.641 earns **P3**.
 - Writing $\int_0^{20} f(t) + 30$ in the presence of an answer of 95, 96, 95.6, or 95.64 earns **P3**.
 - Writing $\int_0^{20} f(t) + 30$ in the presence of no answer or an incorrect answer does not earn **P3**.
- Writing $\int f(t)dt$, $30 + \int f(t)dt$, $30 + \int f(t)$, or $\int f(t) + 30$ and never providing the definite integral, does not earn **P3**.

Model Solution:

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$$30 + \int_0^{20} f(t)dt = 30 + \int_0^{20} (2 \ln(t + 3) - 1.6)dt$$



0	1
---	---

The response accurately includes the criteria below.

- Definite integral

Part B Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P4** is earned for the correct answer, with or without supporting work. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point, unless an earlier point was not earned due to inappropriate rounding.
- Incorrect or unclear communication between the correct integral and the correct answer is treated as scratch work and is not considered in scoring. The following examples earn **P4** for the correct answer:
 - $\int_0^{20} f(t)dt = 65.641060$ so the number of liters of water is 95.641.
 - $\int_0^{20} f(t)dt = 65.641060 + 30 = 95.641060$ (Incorrect linkage is not considered in scoring.)
 - $\int_0^{20} f(t)dt = 95.641060$ (Incorrect linkage is not considered in scoring.)

Model Solution:

There are 95.641 liters of water in the tank at time $t = 20$ minutes.



0	1
---	---

The response accurately includes the criteria below.

- Answer

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

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- To earn **P5**, a response must indicate $\lim_{t \rightarrow 20^+} w(t) = 4.704$ (or 4.703), either symbolically or verbally. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn **P5**, unless an earlier point was not earned due to inappropriate rounding.
 - A response of “the right-hand limit is 4.704” is sufficient to earn **P5**.

Model Solution:

$$\lim_{t \rightarrow 20^+} w(t) = \lim_{t \rightarrow 20^+} g(t) = 4.703562$$

The limit as t approaches 20 from the right is 4.704 (or 4.703).



0	1
---	---

The response accurately includes the criteria below.

- Right-hand limit

Part C Point 6

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P6**, a response must include $w(20) = 4.671$ (or 4.67 or 4.6), $\lim_{t \rightarrow 20^+} w(t) \neq w(20)$, and a conclusion of not continuous (with or without indicating $t = 20$).
- Alternate solution:

$$\lim_{t \rightarrow 20^-} w(t) = \lim_{t \rightarrow 20^-} f(t) = 4.670988$$

Because $\lim_{t \rightarrow 20^-} w(t) \neq \lim_{t \rightarrow 20^+} w(t)$, w is not continuous at $t = 20$.

Such a response is scored analogously to the model solution and earns **P6**.

- Special case: A response of “ $\lim_{t \rightarrow 20^+} w(t) = 4.7$ and $\lim_{t \rightarrow 20^-} w(t) = w(20) = 4.7$, therefore continuous” earns **P6**.

Model Solution:

$$w(20) = f(20) = 4.670988$$

Because $\lim_{t \rightarrow 20^+} w(t) \neq w(20)$, w is not continuous at $t = 20$.

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0	1
---	---

The response accurately includes the criteria below.

- Answer with justification

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P7**, a response must present one or more definite integrals with correct integrands, with or without differentials. Limits of integration are not considered in scoring for **P7**. That is, any of the following earns **P7**: $\int_a^c w(t)dt$, $\int_a^b w(t)dt + \int_b^c w(t)dt$, or $\int_a^b f(t)dt + \int_b^c g(t)dt$, where a , b , and c are real numbers between 0 and 45, inclusive.

Model Solution:

$$\frac{1}{30-15} \int_{15}^{30} w(t)dt$$



0	1
---	---

The response accurately includes the criteria below.

- Definite integral

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P8**, a response must present a correct definite integral and evidence of division by 15.
 - A response that earned **P7** earns **P8** for $\frac{1}{15} \int_{15}^{30} w(t)dt$, $\frac{1}{15} \left(\int_{15}^{20} w(t)dt + \int_{20}^{30} w(t)dt \right)$, $\frac{1}{15} \left(\int_{15}^{20} f(t)dt + \int_{20}^{30} g(t)dt \right)$, or equivalent.

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- Presenting the correct integral and the correct answer will suffice as evidence of division by 15 and will earn **P8**. Work presented between earning **P7** and the correct answer that does not reveal incorrect work done in the calculator will be treated as scratch work and will not be considered in scoring.
 - “ $\int_{15}^{30} w(t)dt = 67.559$, so 4.504” earns **P8** for correct limits of integration and an implied division by 15 made evident by the correct answer.
 - “ $\int_{15}^{30} w(t)dt = \frac{67.559}{15} = 4.504$ ” earns **P8** for correct limits of integration and an implied division by 15 made evident by the correct answer. The incorrect communication between the integral and the answer does not reveal incorrect work done in the calculator. This work is treated as scratch work and is not considered in scoring of **P8**.
- A response that did not earn **P7** is eligible to earn **P8** for $\frac{1}{15} \int_{15}^{30} f(t)dt$ or $\frac{1}{15} \int_{15}^{30} g(t)dt$.

Model Solution:

$$\begin{aligned} & \frac{1}{15} \left(\int_{15}^{20} w(t)dt + \int_{20}^{30} w(t)dt \right) \\ &= \frac{1}{15} \left(\int_{15}^{20} f(t)dt + \int_{20}^{30} g(t)dt \right) \end{aligned}$$



0	1
---	---

The response accurately includes the criteria below.

- Average value formula

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- P9** is earned for the correct answer, with or without supporting work. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point, unless an earlier point was not earned due to inappropriate rounding.

Model Solution:

The average value of w from time $t = 15$ and time $t = 30$ minutes is 4.504 (or 4.503) liters per minute.

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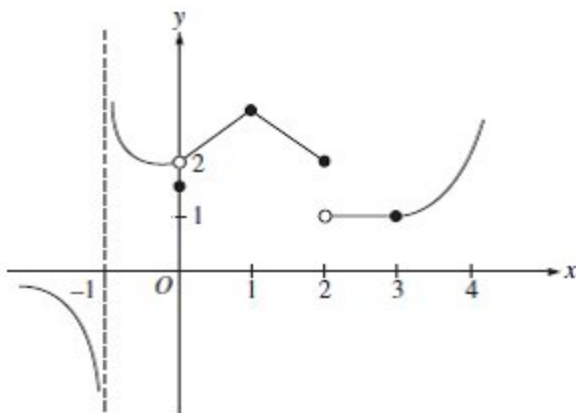
0

1

The response accurately includes the criteria below.

- Answer

29.



The graph of a function f is shown above. If $\lim_{x \rightarrow b} f(x)$ exists and f is not continuous at b , then $b =$

- (A) -1
 (B) 0
 (C) 1
 (D) 2
 (E) 3



30.

Let f be the function given by $f(x) = \frac{x-2}{2|x-2|}$. Which of the following is true?

- (A) $\lim_{x \rightarrow 2} f(x) = \frac{1}{2}$
 (B) f has a removable discontinuity at $x = 2$.
 (C) f has a jump discontinuity at $x = 2$.
 (D) f has a discontinuity due to a vertical asymptote at $x = 2$.



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31. Let f be the function given by $f(x) = \ln(x / x - 1)$.
- (a) What is the domain of f ?
- (b) Find the value of the derivative of f at $x = -1$.
- (c) Write an expression for $f^{-1}(x)$, where f^{-1} denotes the inverse function of f .

Part A

1 point is earned for setting $\frac{x}{x-1} > 0$

1 point is earned for first interior point

1 point is earned for second interior point

1 point is **deducted** for each incorrect point

$$\frac{x}{x-1} > 0$$

$$x > 0 \text{ and } x - 1 > 0 \Rightarrow x > 1$$

$$x < 0 \text{ and } x - 1 < 0 \Rightarrow x < 0$$

$$x < 0 \text{ or } x > 1$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for setting $\frac{x}{x-1} > 0$

1 point is earned for first interior point

1 point is earned for second interior point

1 point is **deducted** for each incorrect point

$$\frac{x}{x-1} > 0$$

$$x > 0 \text{ and } x - 1 > 0 \Rightarrow x > 1$$

$$x < 0 \text{ and } x - 1 < 0 \Rightarrow x < 0$$

$$x < 0 \text{ or } x > 1$$

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Part B

2 point is earned for finding the derivative

2 point is **deducted** for error in chain, product, quotient or derivative of $\ln$.

1 point is **deducted** for all other error

1 point is earned for evaluating his/her $f'(x)$ at $x = -1$

$$f'(x) = \frac{x-1}{x} \cdot \frac{(x-1)-x}{(x-1)^2}$$

$$= \frac{-1}{x(x-1)}$$

or

$$\ln|x| - \ln|x-1| \Rightarrow f'(x) = \frac{1}{x} - \frac{1}{x-1}$$

$$f'(-1) = -\frac{1}{2}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 point is earned for finding the derivative

2 point is **deducted** for error in chain, product, quotient or derivative of $\ln$.

1 point is **deducted** for all other error

1 point is earned for evaluating his/her $f'(x)$ at $x = -1$

$$f'(x) = \frac{x-1}{x} \cdot \frac{(x-1)-x}{(x-1)^2}$$

$$= \frac{-1}{x(x-1)}$$

or

$$\ln|x| - \ln|x-1| \Rightarrow f'(x) = \frac{1}{x} - \frac{1}{x-1}$$

$$f'(-1) = -\frac{1}{2}$$

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Part C

1 point is earned for correctly introducing exponential

1 point is earned for solving for x in terms of e^y

1 point is earned for expressing solution in terms of x

$$y = \ln\left(\frac{x}{x-1}\right)$$

$$e^y = \frac{x}{x-1}$$

$$x(e^y - 1) = e^y$$

$$x = \frac{e^y}{e^y - 1}$$

$$f^{-1}(x) = \frac{e^x}{e^x - 1}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for correctly introducing exponential

1 point is earned for solving for x in terms of e^y

1 point is earned for expressing solution in terms of x

$$y = \ln\left(\frac{x}{x-1}\right)$$

$$e^y = \frac{x}{x-1}$$

$$x(e^y - 1) = e^y$$

$$x = \frac{e^y}{e^y - 1}$$

$$f^{-1}(x) = \frac{e^x}{e^x - 1}$$

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32. Let f be the function given by $f(x) = \frac{x}{\sqrt{x^2 - 4}}$.

- (a) Find the domain of f .
- (b) Write an equation for each vertical asymptote to the graph of f .
- (c) Write an equation for each horizontal asymptote to the graph of f .
- (d) Find $f'(x)$.

Part A

1 point is earned for $x^2 - 4 > 0$

1 point is earned for correct solution

$$x < -2 \text{ or } x > 2$$

$$\text{OR } |x| > 2$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $x^2 - 4 > 0$

1 point is earned for correct solution

$$x < -2 \text{ or } x > 2$$

$$\text{OR } |x| > 2$$

Part B

1 point is earned for finding value of x

$$x = 2, x = -2$$

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0	1
---	---

The student response earns all of the following points:

1 point is earned for finding value of x

$$x = 2, x = -2$$

Part C

1 point is earned for analysis in both direction

1 point is earned for 1st consistent solution

1 point is earned for 2nd consistent solution

$$\lim_{x \rightarrow +\infty} \frac{x}{\sqrt{x^2 - 4}} = 1$$

$$\lim_{x \rightarrow -\infty} \frac{1}{\sqrt{x^2 - 4}} = -1$$

$$y = 1, y = -1$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for analysis in both direction

1 point is earned for 1st consistent solution

1 point is earned for 2nd consistent solution

$$\lim_{x \rightarrow +\infty} \frac{x}{\sqrt{x^2 - 4}} = 1$$

$$\lim_{x \rightarrow -\infty} \frac{1}{\sqrt{x^2 - 4}} = -1$$

$$y = 1, y = -1$$

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Part D

3 points are earned for finding $f'(x)$

2 points are **deducted** for Chain rule error, Quotient (Product) rule error; and calculus error in exponent

1 point is **deducted** for all other errors

$$\begin{aligned}
 f'(x) &= \frac{\sqrt{x^2-4} - x \left[\frac{1}{2}(x^2-4)^{-1/2} 2x \right]}{x^2-4} \\
 &= \frac{\sqrt{x^2-4} - \frac{x^2}{\sqrt{x^2-4}}}{x^2-4} \\
 &= \frac{-4}{(x^2-4)^{3/2}}
 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

3 points are earned for finding $f'(x)$

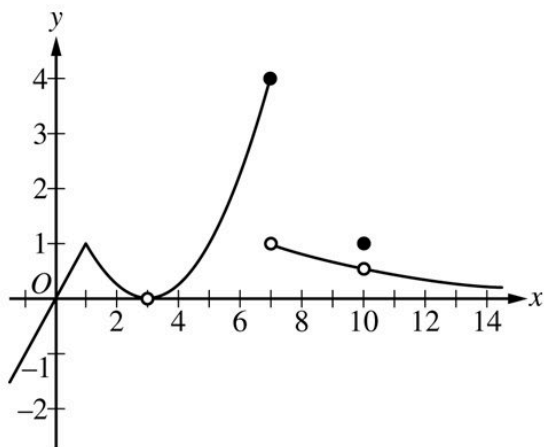
2 points are **deducted** for Chain rule error, Quotient (Product) rule error; and calculus error in exponent

1 point is **deducted** for all other errors

$$\begin{aligned}
 f'(x) &= \frac{\sqrt{x^2-4} - x \left[\frac{1}{2}(x^2-4)^{-1/2} 2x \right]}{x^2-4} \\
 &= \frac{\sqrt{x^2-4} - \frac{x^2}{\sqrt{x^2-4}}}{x^2-4} \\
 &= \frac{-4}{(x^2-4)^{3/2}}
 \end{aligned}$$

Unit 1 AP Topics 9-12

33.

Graph of f

The graph of the function f is shown above. At what value of x does f have a jump discontinuity?

- (A) 1
(B) 3
(C) 7
(D) 10

**Answer C**

This option is correct. A jump discontinuity occurs at $x = c$ if the left- and right-sided limits exist at c , but are not equal. that is,

$$\lim_{x \rightarrow c^-} f(x) \neq \lim_{x \rightarrow c^+} f(x).$$

For this function,

$$\lim_{x \rightarrow 7^-} f(x) = 4$$

$$\lim_{x \rightarrow 7^+} f(x) = 1$$

and so there is a jump discontinuity at $x = 7$.

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34.

Graph of f

The graph of a function f is shown in the figure above. At what value of x does f have a removable discontinuity?

- (A) $x = 1$
(B) $x = 3$
(C) $x = 4$
(D) $x = 5$

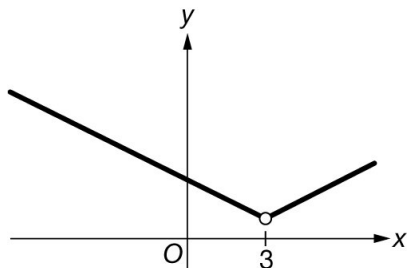
**Answer B**

Correct. f has a removable discontinuity at $x = 3$, because $\lim_{x \rightarrow 3} f(x)$ exists and is not equal to $f(3)$. That is, $\lim_{x \rightarrow 3} f(x) = 0 \neq 1 = f(3)$.

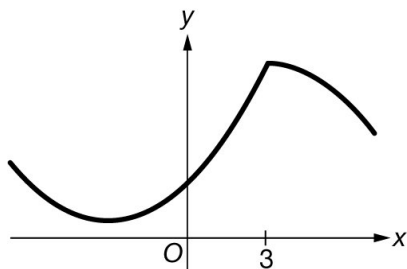
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35. The function f has a jump discontinuity at $x = 3$. Which of the following could be the graph of f ?

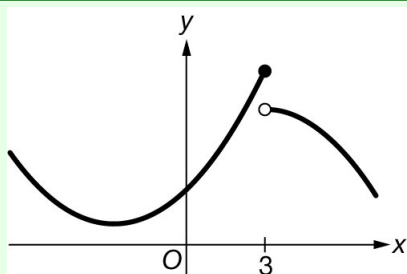
(A)



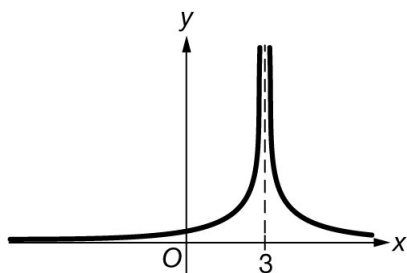
(B)



(C)



(D)

**Answer C**

Correct. This graph has a jump discontinuity at $x = 3$, because $\lim_{x \rightarrow 3^-} f(x)$ and $\lim_{x \rightarrow 3^+} f(x)$ are finite but not equal to each other.

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36.

$$f(x) = \begin{cases} \frac{2x^2-3x-2}{x-2} & \text{if } x \neq 2 \\ 7 & \text{if } x = 2 \end{cases}$$

The function f is defined above. Which of the following statements is true?

- (A) f is continuous at $x = 2$.
- (B) f has a removable discontinuity at $x = 2$. ✓
- (C) f has a jump discontinuity at $x = 2$.
- (D) f has a discontinuity due to a vertical asymptote at $x = 2$.

Answer B

Correct. $\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{2x^2-3x-2}{x-2} = \lim_{x \rightarrow 2} \frac{(2x+1)(x-2)}{x-2} = \lim_{x \rightarrow 2} (2x+1) = 2 \cdot 2 + 1 = 5$
 $\lim_{x \rightarrow 2} f(x) = 5$, but $f(2) \neq 5$, so there is a removable discontinuity at $x = 2$.

37. Which of the following functions is continuous at $x = 3$?

(A) $f(x) = \begin{cases} \frac{x^2-x-6}{x-3} & \text{for } x \neq 3 \\ 8 & \text{for } x = 3 \end{cases}$

(B) $g(x) = \begin{cases} 5 & \text{for } x < 3 \\ 3x - 4 & \text{for } x > 3 \end{cases}$

(C) $h(x) = \begin{cases} \sin\left(\frac{\pi}{2}x\right) & \text{for } x < 3 \\ -1 & \text{for } x = 3 \\ \cos(\pi x) & \text{for } x > 3 \end{cases}$ ✓

(D) $k(x) = \begin{cases} 5 + \ln(4-x) & \text{for } x \leq 3 \\ 5 \ln(x-2) & \text{for } x > 3 \end{cases}$

Answer C

Correct. The function h is continuous at $x = 3$.

$h(3)$ exists and $h(3) = -1$.

$$\lim_{x \rightarrow 3^-} h(x) = \lim_{x \rightarrow 3^-} \left(\sin\left(\frac{\pi}{2}x\right) \right) = \sin\left(\frac{3\pi}{2}\right) = -1$$

$$\lim_{x \rightarrow 3^+} h(x) = \lim_{x \rightarrow 3^+} (\cos(\pi x)) = \cos(3\pi) = -1$$

Therefore, $\lim_{x \rightarrow 3} h(x)$ exists and $\lim_{x \rightarrow 3} h(x) = -1 = h(3)$.

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38.

$$f(x) = \begin{cases} 3^x & \text{for } 0 < x < 1 \\ \frac{1}{2}x^2 - x + \frac{7}{2} & \text{for } 1 < x < 2 \end{cases}$$

Let f be the function defined above. Which of the following statements is true?

(A) f is continuous at $x = 1$.

(B) f is not continuous at $x = 1$ because $f(1)$ does not exist. ✓

(C) f is not continuous at $x = 1$ because $\lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x)$.

(D) f is not continuous at $x = 1$ because $\lim_{x \rightarrow 1} f(x)$ does not exist.

Answer B

Correct. In order for f to be continuous at $x = 1$, the following conditions must be satisfied: $f(1)$ exists, $\lim_{x \rightarrow 1} f(x)$ exists, and $\lim_{x \rightarrow 1} f(x) = f(1)$. However, the function f is defined only for $0 < x < 1$ and $1 < x < 2$. Therefore, $f(1)$ does not exist.

39. If $\lim_{x \rightarrow 6} f(x)$ exists with $\lim_{x \rightarrow 6} f(x) < 8$ and $f(6) = 12$, which of the following statements must be false?

(A) $\lim_{x \rightarrow 6^-} f(x) = 0$

(B) $\lim_{x \rightarrow 6^+} f(x) < 8$

(C) $\lim_{x \rightarrow 6^-} f(x) = \lim_{x \rightarrow 6^+} f(x)$

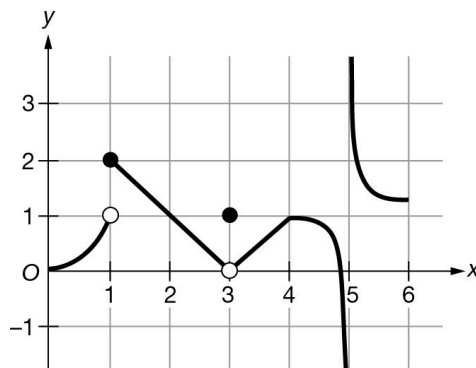
(D) f is continuous at $x = 6$. ✓

Answer D

Correct. Since $\lim_{x \rightarrow 6} f(x) < 8$ and $f(6) = 12$, it is false that $\lim_{x \rightarrow 6} f(x) = f(6)$. Therefore, f cannot be continuous at $x = 6$.

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40.

Graph of f

The graph of a function f is shown in the figure above. At what value of x does f have a jump discontinuity?

(A) $x = 1$



(B) $x = 3$

(C) $x = 4$

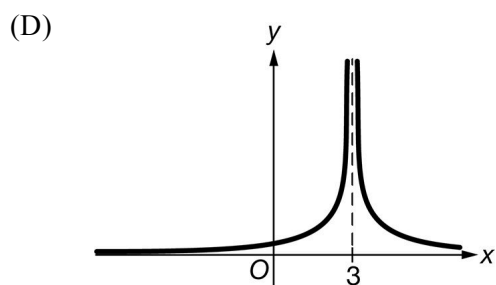
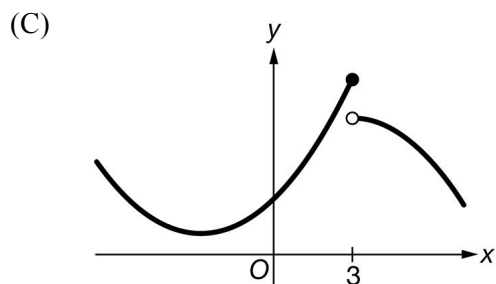
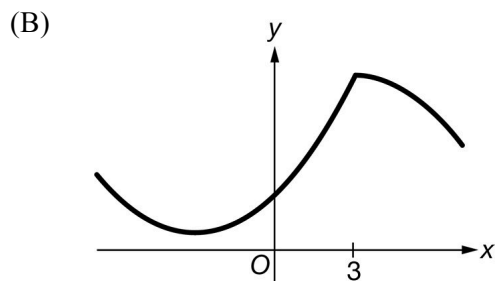
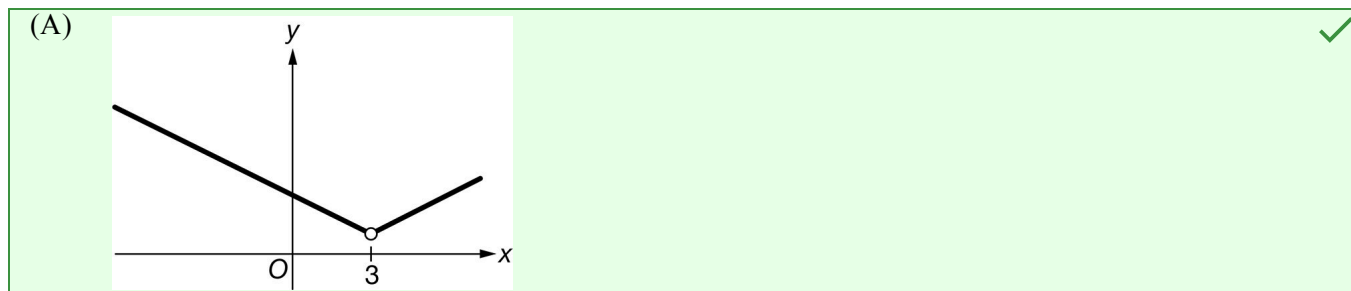
(D) $x = 5$

Answer A

Correct. f has a jump discontinuity at $x = 1$, because $\lim_{x \rightarrow 1^-} f(x)$ and $\lim_{x \rightarrow 1^+} f(x)$ exist but are not equal to each other. That is, $\lim_{x \rightarrow 1^-} f(x) = 1$ and $\lim_{x \rightarrow 1^+} f(x) = 2$, and therefore, $\lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x)$.

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41. The function f has a removable discontinuity at $x = 3$. Which of the following could be the graph of f ?

**Answer A**

Correct. This graph has a removable discontinuity at $x = 3$, because $\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x)$, even though the function is not defined at $x = 3$.

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42.

$$f(x) = \begin{cases} \frac{2x^2-5x-3}{x-3} & \text{if } x \neq 3 \\ 9 & \text{if } x = 3 \end{cases}$$

The function f is defined above. Which of the following statements is true?

- (A) f is continuous at $x = 3$.
- (B) f has a removable discontinuity at $x = 3$. ✓
- (C) f has a jump discontinuity at $x = 3$.
- (D) f has a discontinuity due to a vertical asymptote at $x = 3$.

Answer B

Correct. $\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} \frac{2x^2-5x-3}{x-3} = \lim_{x \rightarrow 3} \frac{(2x+1)(x-3)}{x-3} = \lim_{x \rightarrow 3} (2x+1) = 2 \cdot 3 + 1 = 7$
 $\lim_{x \rightarrow 3} f(x) = 7$, but $f(3) \neq 7$, so there is a removable discontinuity at $x = 3$.

43.

$$f(x) = \begin{cases} 2x + 3 & \text{for } x < 1 \\ 5 & \text{for } x = 1 \\ -4x + 9 & \text{for } 1 < x < 3 \\ 4 & \text{for } x = 3 \\ x - 6 & \text{for } x > 3 \end{cases}$$

Let f be the piecewise function defined above. Which of the following statements is false?

- (A) f is continuous at $x = 1$.
- (B) f is continuous at $x = 2$.
- (C) f is continuous at $x = 3$. ✓
- (D) f is continuous at $x = 4$.

Answer C

Correct. The statement is false because $\lim_{x \rightarrow 3} f(x) = -3$ but $f(3) = 4$. In order for f to be continuous at $x = 3$, $\lim_{x \rightarrow 3} f(x)$ must equal $f(3)$.

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44. A student attempted to confirm that the function f defined by $f(x) = \frac{x^2-5x+4}{x^2-6x+8}$ is continuous at $x = 4$. In which step, if any, does an error first appear?

Step 1: $f(x) = \frac{x^2-5x+4}{x^2-6x+8} = \frac{(x-4)(x-1)}{(x-4)(x-2)}$

Step 2: $\lim_{x \rightarrow 4} f(x) = \lim_{x \rightarrow 4} \frac{x-1}{x-2} = \frac{3}{2}$

Step 3: $f(4) = \frac{4-1}{4-2} = \frac{3}{2}$

Step 4: $\lim_{x \rightarrow 4} f(x) = f(4)$, so f is continuous at $x = 4$.

(A) Step 2

(B) Step 3

(C) Step 4

(D) There is no error in the confirmation.



Answer B

Correct. $f(4)$ does not exist, because 4 is not in the domain of f . The common term $x - 4$ cannot be canceled from the numerator and denominator when $x = 4$.

45.

$$f(x) = \begin{cases} 2^x & \text{for } 0 < x < 1 \\ \frac{1}{2}x^2 - x + \frac{5}{2} & \text{for } 1 < x < 2 \end{cases}$$

Let f be the function defined above. Which of the following statements is true?

(A) f is continuous at $x = 1$.

(B) f is not continuous at $x = 1$ because $f(1)$ does not exist.

(C) f is not continuous at $x = 1$ because $\lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x)$.

(D) f is not continuous at $x = 1$ because $\lim_{x \rightarrow 1} f(x)$ does not exist.



Answer B

Correct. In order for f to be continuous at $x = 1$, the following conditions must be satisfied: $f(1)$ exists, $\lim_{x \rightarrow 1} f(x)$ exists, and $\lim_{x \rightarrow 1} f(x) = f(1)$. However, the function f is defined only for $0 < x < 1$ and $1 < x < 2$. Therefore, $f(1)$ does not exist.

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46. If $\lim_{x \rightarrow 6} f(x)$ exists with $\lim_{x \rightarrow 6} f(x) < 5$ and $f(6) = 10$, which of the following statements must be false?

- (A) $\lim_{x \rightarrow 6^-} f(x) = 0$
(B) $\lim_{x \rightarrow 6^+} f(x) < 5$
(C) $\lim_{x \rightarrow 6^-} f(x) = \lim_{x \rightarrow 6^+} f(x)$
(D) f is continuous at $x = 6$.

**Answer D**

Correct. Since $\lim_{x \rightarrow 6} f(x) < 5 < 10 = f(6)$, it is false that $\lim_{x \rightarrow 6} f(x) = f(6)$. Therefore f cannot be continuous at $x = 6$.

47. Let f be the function defined by $f(x) = \frac{3x^3+2x^2}{x^2-x}$. Which of the following statements is true?

- (A) f has a discontinuity due to a vertical asymptote at $x = 0$ and at $x = 1$.
(B) f has a removable discontinuity at $x = 0$ and a jump discontinuity at $x = 1$.
(C) f has a removable discontinuity at $x = 0$ and a discontinuity due to a vertical asymptote at $x = 1$.
(D) f is continuous at $x = 0$, and f has a discontinuity due to a vertical asymptote at $x = 1$.

**Answer C**

Correct. The function f is not defined at $x = 0$ because the denominator equals 0 when $x = 0$. However, $\lim_{x \rightarrow 0} f(x)$ exists, as shown below. Therefore, f has a removable discontinuity at $x = 0$.

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{3x^3+2x^2}{x^2-x} = \lim_{x \rightarrow 0} \frac{x(3x^2+2x)}{x(x-1)} = \lim_{x \rightarrow 0} \frac{3x^2+2x}{x-1} = \frac{3 \cdot 0^2 + 2 \cdot 0}{0-1} = 0$$

The graph of the rational function f has a vertical asymptote at $x = 1$ because the numerator is nonzero and the denominator equals 0 when $x = 1$.

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48.  Let f be the function defined by $f(x) = \frac{x^3 - 2x^2 - 3x}{x^3 - 3x^2 + 4}$. Which of the following statements about f at $x = 2$ and $x = -1$ is true?

- (A) f has a jump discontinuity at $x = 2$, and f is continuous at $x = -1$.
- (B) f has a jump discontinuity at $x = 2$, and f has a removable discontinuity at $x = -1$.
- (C) f has a discontinuity due to a vertical asymptote at $x = 2$, and f is continuous at $x = -1$.
- (D) f has a discontinuity due to a vertical asymptote at $x = 2$, and f has a removable discontinuity at $x = -1$. 

Answer D

Correct. A graphing calculator is used to graph the function f . The graph suggests that there is a discontinuity due to a vertical asymptote at $x = 2$. To verify this, let $g(x) = x^3 - 2x^2 - 3x$ and $h(x) = x^3 - 3x^2 + 4$ so that $f(x) = \frac{g(x)}{h(x)}$.

There is a discontinuity due to a vertical asymptote at $x = 2$ because $g(2) = -6 \neq 0$ and $h(2) = 0$. The graph might appear to be continuous at $x = -1$. However, there is a removable discontinuity at $x = -1$. $\lim_{x \rightarrow -1} f(x)$ does exist, but $f(-1)$ is not defined because $g(-1) = h(-1) = 0$.

49.  $f(x) = \frac{x^3 + 4x^2 + x - 6}{3 \sin(-\frac{\pi}{2}x) + 3x^2}$

Let f be the function defined above. Which of the following conditions explains why f is not continuous at $x = 1$?

- (A) Neither $\lim_{x \rightarrow 1} f(x)$ nor $f(1)$ exists.
- (B) $\lim_{x \rightarrow 1} f(x)$ exists, but $f(1)$ does not exist. 
- (C) Both $\lim_{x \rightarrow 1} f(x)$ and $f(1)$ exist, but $\lim_{x \rightarrow 1} f(x) \neq f(1)$.
- (D) Both $\lim_{x \rightarrow 1} f(x)$ and $f(1)$ exist, and $\lim_{x \rightarrow 1} f(x) = f(1)$.

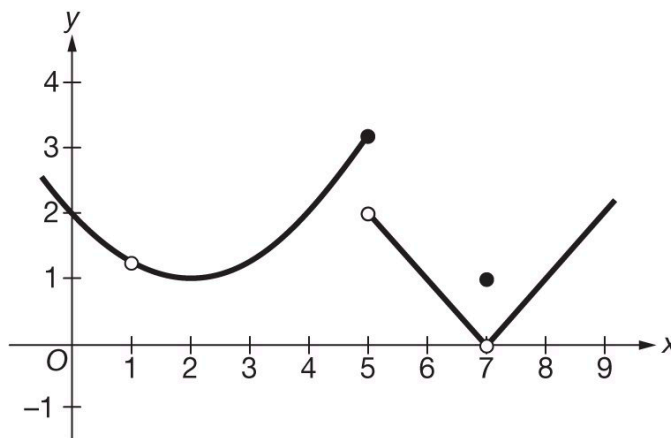
Answer B

Correct. The calculator is used to graph f near $x = 1$. The graph suggests that $\lim_{x \rightarrow 1} f(x)$ exists. The function f is not defined at $x = 1$, however, since the denominator of the expression for $f(x)$ is equal to 0 at $x = 1$.

$$3 \sin\left(-\frac{\pi}{2} \cdot 1\right) + 3(1)^2 = 3 \sin\left(-\frac{\pi}{2}\right) + 3 = 0$$

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50.

Graph of f

The graph of the function f is shown above. What are all values of x for which f has a removable discontinuity?

- (A) 1 only
(B) 5 only
(C) 1 and 7 only
(D) 1, 5, and 7

**Answer C**

Correct. A removable discontinuity occurs at $x = c$ if $\lim_{x \rightarrow c} f(x) = L$ exists but $f(c)$ is not defined or $f(c) \neq L$.

For this function, $\lim_{x \rightarrow 1} f(x)$ exists but $f(1)$ is not defined, and $\lim_{x \rightarrow 7} f(x)$ exists but $f(7) \neq \lim_{x \rightarrow 7} f(x) = 0$.

Therefore, there are removable discontinuities at $x = 1$ and at $x = 7$.

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51.

$$f(x) = \frac{x^3 + 2x - 12}{8 \cos\left(\frac{\pi}{2}x\right) + 2x^2}$$

Let f be the function defined above. Which of the following conditions explains why f is not continuous at $x = 2$?

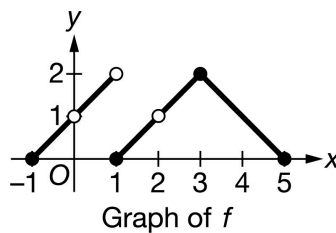
- (A) Neither $\lim_{x \rightarrow 2} f(x)$ nor $f(2)$ exists.
- (B) $\lim_{x \rightarrow 2} f(x)$ exists, but $f(2)$ does not exist. ✓
- (C) Both $\lim_{x \rightarrow 2} f(x)$ and $f(2)$ exist, but $\lim_{x \rightarrow 2} f(x) \neq f(2)$.
- (D) Both $\lim_{x \rightarrow 2} f(x)$ and $f(2)$ exist, and $\lim_{x \rightarrow 2} f(x) = f(2)$.

Answer B

Correct. The calculator is used to graph f near $x = 2$. The graph suggests that $\lim_{x \rightarrow 2} f(x)$ exists. The function f is not defined at $x = 2$, however, since the denominator of the expression for $f(x)$ is equal to 0 at $x = 2$.

$$8 \cos\left(\frac{\pi}{2} \cdot 2\right) + 2(2)^2 = 8 \cos \pi + 8 = 0$$

52.



The graph of the function f is shown above. What are all values of x for which f has a removable discontinuity?

- (A) 0 only
- (B) 1 only
- (C) 0 and 2 only ✓
- (D) 0, 1, and 2

Answer C

Correct. A removable discontinuity occurs at $x = c$ if $\lim_{x \rightarrow c} f(x) = L$ exists but $f(c)$ is not defined or $f(c) \neq L$.

For this function, $\lim_{x \rightarrow 0} f(x)$ exists but $f(0)$ is not defined, and $\lim_{x \rightarrow 2} f(x)$ exists but $f(2)$ is not defined. Therefore, there are removable discontinuities at $x = 0$ and at $x = 2$.

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53. Let f be the function defined by $f(x) = \frac{x^4 - 4x^2}{x^2 - 4x}$. Which of the following statements is true?

- (A) f has a discontinuity due to a vertical asymptote at $x = 0$ and at $x = 4$.
 (B) f has a removable discontinuity at $x = 0$ and a jump discontinuity at $x = 4$.
 (C) f has a removable discontinuity at $x = 0$ and a discontinuity due to a vertical asymptote at $x = 4$. ✓
 (D) f is continuous at $x = 0$, and f has a discontinuity due to a vertical asymptote at $x = 4$.

Answer C

Correct. The function f is not defined at $x = 0$ because the denominator equals 0 when $x = 0$. However, $\lim_{x \rightarrow 0} f(x)$ exists, as shown below. Therefore, f has a removable discontinuity at $x = 0$.

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{x^4 - 4x^2}{x^2 - 4x} = \lim_{x \rightarrow 0} \frac{x^2(x^2 - 4)}{x(x - 4)} = \lim_{x \rightarrow 0} \frac{x(x+2)(x-2)}{x-4} = \frac{0(0+2)(0-2)}{(0-4)} = 0$$

The graph of the rational function f has a vertical asymptote at $x = 4$ because the numerator is nonzero and the denominator equals 0 when $x = 4$.

54. A student attempted to confirm that the function f defined by $f(x) = \frac{x^2 + x - 6}{x^2 - 7x + 10}$ is continuous at $x = 2$. In which step, if any, does an error first appear?

- Step 1: $f(x) = \frac{x^2 + x - 6}{x^2 - 7x + 10} = \frac{(x-2)(x+3)}{(x-2)(x-5)}$
 Step 2: $\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{x+3}{x-5} = \frac{2+3}{2-5} = -\frac{5}{3}$
 Step 3: $f(2) = \frac{2+3}{2-5} = -\frac{5}{3}$
 Step 4: $\lim_{x \rightarrow 2} f(x) = f(2)$, so f is continuous at $x = 2$.

- (A) Step 2
 (B) Step 3 ✓
 (C) Step 4
 (D) There is no error in the confirmation.

Answer B

Correct. $f(2)$ does not exist, because the denominator of the expression is equal to 0 when $x = 2$, so 2 is not in the domain of f . The common term $x - 2$ cannot be canceled from the numerator and denominator when evaluating the function at $x = 2$.

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55.

$$f(x) = \begin{cases} x^2 + 2x & \text{for } x < 1 \\ 3 & \text{for } x = 1 \\ x^3 + x^2 + x & \text{for } 1 < x < 3 \\ 0 & \text{for } x = 3 \\ 2x + 1 & \text{for } x > 3 \end{cases}$$

Let f be the piecewise function defined above. Which of the following statements is false?

- (A) f is continuous at $x = 1$.
(B) f is continuous at $x = 2$.
(C) f is continuous at $x = 3$.
(D) f is continuous at $x = 4$.

**Answer C**

Correct. The statement is false because $\lim_{x \rightarrow 3^-} f(x) = 27 + 9 + 3 = 39 \neq f(3)$. In order for f to be continuous at $x = 3$, $\lim_{x \rightarrow 3} f(x)$ must exist and be equal to 3.

56.  Let f be the function defined by $f(x) = \frac{x^3 - 9x}{x^3 + x^2 - 8x - 12}$. Which of the following statements about f at $x = -2$ and $x = 3$ is true?

- (A) f has a jump discontinuity at $x = -2$, and f is continuous at $x = 3$.
(B) f has a jump discontinuity at $x = -2$, and f has a removable discontinuity at $x = 3$.
(C) f has a discontinuity due to a vertical asymptote at $x = -2$, and f is continuous at $x = 3$.
(D) f has a discontinuity due to a vertical asymptote at $x = -2$, and f has a removable discontinuity at $x = 3$.

**Answer D**

Correct. A graphing calculator is used to graph the function f . The graph suggests that there is a discontinuity due to a vertical asymptote at $x = -2$. To verify this, let $g(x) = x^3 - 9x$ and $h(x) = x^3 + x^2 - 8x - 12$ so that $f(x) = \frac{g(x)}{h(x)}$. There is a discontinuity due to a vertical asymptote at $x = -2$ because $g(-2) = 10 \neq 0$ and $h(-2) = 0$. The graph might appear to be continuous at $x = 3$. However, there is a removable discontinuity at $x = 3$. $\lim_{x \rightarrow 3} f(x)$ does exist, but $f(3)$ is not defined because $g(3) = h(3) = 0$.

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57. Which of the following functions is continuous at $x = 3$?

(A)
$$f(x) = \begin{cases} \frac{x^2+x-12}{x-3} & \text{for } x \neq 3 \\ 8 & \text{for } x = 3 \end{cases}$$

(B)
$$g(x) = \begin{cases} 4x - 7 & \text{for } x < 3 \\ \text{undefined} & \text{for } x = 3 \\ x + 2 & \text{for } x > 3 \end{cases}$$

(C)
$$h(x) = \begin{cases} -8 \sin\left(\frac{\pi}{2}x\right) & \text{for } x < 3 \\ 8 & \text{for } x = 3 \\ -8 \cos(\pi x) & \text{for } x > 3 \end{cases}$$
 ✓

(D)
$$k(x) = \begin{cases} 8 + \ln(4 - x) & \text{for } x \leq 3 \\ 8 \ln(x - 2) & \text{for } x > 3 \end{cases}$$

Answer C

Correct. The function h is continuous at $x = 3$.

$h(3)$ exists and $h(3) = 8$.

$$\lim_{x \rightarrow 3^-} h(x) = \lim_{x \rightarrow 3^-} (-8 \sin(\frac{\pi}{2}x)) = -8 \sin(\frac{3\pi}{2}) = 8$$

$$\lim_{x \rightarrow 3^+} h(x) = \lim_{x \rightarrow 3^+} (-8 \cos(\pi x)) = -8 \cos(3\pi) = 8$$

Therefore $\lim_{x \rightarrow 3} h(x)$ exists and $\lim_{x \rightarrow 3} h(x) = 8 = h(3)$.

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58.

The function f is continuous at $x = 3$.

$$f(3) = 1 + \lim_{x \rightarrow 3^-} f(x) + \lim_{x \rightarrow 3^+} f(x)$$

Based on the statements above, which of the following must be true?

(A) $f(3) = -1$

(B) $f(3) = 1$

(C) $f(3) \neq \lim_{x \rightarrow 3} f(x)$

(D) f is differentiable at $x = 3$.

**Answer A**

Correct. This statement is true. Because the function f is continuous at $x = 3$, $\lim_{x \rightarrow 3} f(x)$ exists and equals $f(3)$.

Therefore, $\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3^+} f(x) = f(3)$, and so

$$f(3) = 1 + \lim_{x \rightarrow 3^-} f(x) + \lim_{x \rightarrow 3^+} f(x) = 1 + f(3) + f(3) = 1 + 2f(3). \text{ Therefore, } f(3) = -1.$$

59. If the function f is continuous at $x = 3$, which of the following must be true?

(A) $f(3) < \lim_{x \rightarrow 3} f(x)$

(B) $\lim_{x \rightarrow 3^-} f(x) \neq \lim_{x \rightarrow 3^+} f(x)$

(C) $f(3) = \lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x)$

(D) The derivative of f at $x = 3$ exists.

(E) The derivative of f is positive for $x < 3$ and negative for $x > 3$.



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60.

x	$x < 0$	$x = 0$	$0 < x < 1$	$x = 1$	$1 < x < 2$	$x = 2$	$2 < x < 3$	$x = 3$	$3 < x$
$f(x)$	$x^2 - 1$	1	$x^2 + 1$	1	$3 - x$	1	$\frac{1}{x-2}$	1	$\frac{x^2}{9}$

Let f be the piecewise function defined in the table shown. At which of the following values of x does f have a removable discontinuity?

(A) 0

(B) 1

(C) 2

(D) 3

**Answer B**

Correct. A removable discontinuity occurs at $x = c$ if $\lim_{x \rightarrow c} f(x)$ exists but $f(c)$ does not exist or is not equal to the value of the limit. For this function, $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x^2 + 1) = 2$ and $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (3 - x) = 2$, so $\lim_{x \rightarrow 1} f(x)$ exists and equals 2. However, $f(1) = 1$, so f has a removable discontinuity at $x = 1$.

61.

$$g(x) = \begin{cases} 3x^2 - 5x + 1 & \text{if } x \leq 0 \\ \frac{1}{x(x+1)} & \text{if } x > 0 \end{cases}$$

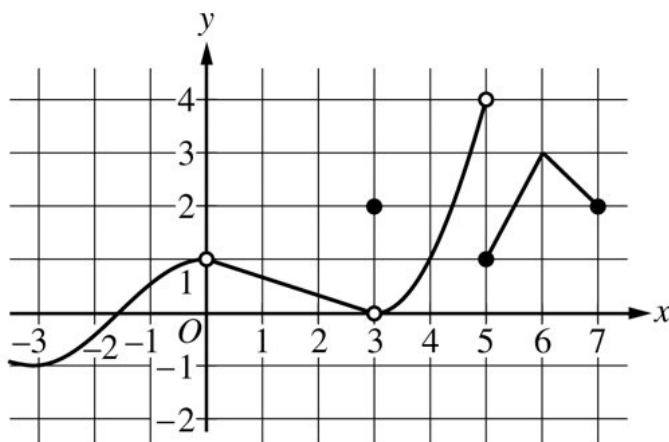
Let g be the function given above. Which of the following statements is true?

(A) g is continuous at $x = 0$.(B) g has a jump discontinuity at $x = 0$.(C) g has a removable discontinuity at $x = 0$.(D) g has a discontinuity due to a vertical asymptote at $x = 0$.**Answer D**

Correct. Because $\lim_{x \rightarrow 0^-} g(x) = \lim_{x \rightarrow 0^-} (3x^2 - 5x + 1) = 1$ but $\lim_{x \rightarrow 0^+} g(x) = \lim_{x \rightarrow 0^+} \frac{1}{x(x+1)} = \infty$, the function g is discontinuous at $x = 0$ and the graph approaches a vertical asymptote as x approaches 0 from the right. Therefore, this statement is true.

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62.

Graph of f

The graph of the function f is shown above. Which of the following statements is true?

- (A) f is discontinuous at $x = 0$ because $\lim_{x \rightarrow 0} f(x)$ does not exist.
- (B) f is discontinuous at $x = 3$ because $\lim_{x \rightarrow 3} f(x) \neq f(3)$. ✓
- (C) f is discontinuous at $x = 5$ because $\lim_{x \rightarrow 5^-} f(x)$ does not exist.
- (D) f is discontinuous at $x = 6$ because $\lim_{x \rightarrow 6^-} f'(x) \neq \lim_{x \rightarrow 6^+} f'(x)$.

Answer B

This option is correct. A function f is continuous at a point $x = a$ if f is defined at $x = a$ and $\lim_{x \rightarrow a} f(x) = f(a)$. Here the function f is defined at $x = 3$ with $f(3) = 2$, but $\lim_{x \rightarrow 3} f(x) = 0 \neq f(3)$ so this statement is true.

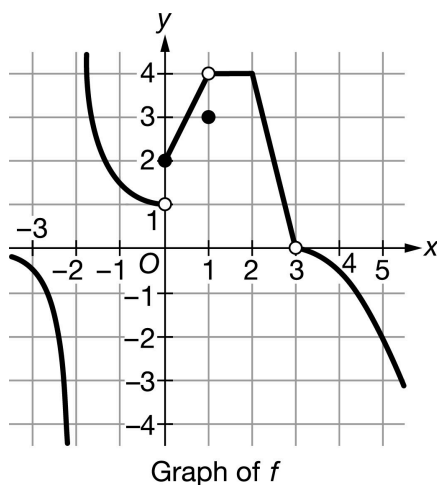
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63. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



x	0.8	0.9	0.99	0.999	1.001	1.01	1.1	1.2
$g(x)$	4.16	4.59	4.960	4.996	5.004	5.040	5.39	5.76

The graph of the function f is shown in the xy -plane above. The graph of f has a vertical asymptote at $x = -2$.

The function g is continuous and increasing for all x . Values of $g(x)$ at selected values of x are shown in the table above.

- Using the graph of f and the table for g , estimate $\lim_{x \rightarrow 1} (2f(x) + 3g(x))$.
- For each of the values $a = -2$, $a = 2$, and $a = 3$, determine whether or not f is continuous at $x = a$. In each case, use the definition of continuity to justify your answer.
- Find the value of $\lim_{x \rightarrow 0} f(f(x))$ or explain why the limit does not exist.
- Write a difference quotient that best approximates the instantaneous rate of change of g at $x = 1$.

Part A

Both limits must be correct to earn the answer point. The response may still earn the points with use of equals rather than approximately. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1	2
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The student response accurately includes both of the criteria below.

- ☐ estimate for $\lim_{x \rightarrow 1} g(x)$
- ☐ answer

Solution:

$$\lim_{x \rightarrow 1} f(x) = 4$$

$$\lim_{x \rightarrow 1} g(x) \approx 5$$

$$\lim_{x \rightarrow 1} (2f(x) + 3g(x)) = 2\lim_{x \rightarrow 1} f(x) + 3\lim_{x \rightarrow 1} g(x) \approx 2 \cdot 4 + 3 \cdot 5 = 23$$

Part B

At most 2 out of 3 points if mathematical notation for any limit is missing or incorrect. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ not continuous at $x = -2$ with reason
- ☐ continuous at $x = 2$ with reason
- ☐ not continuous at $x = 3$ with reason

Solution:

$$\lim_{x \rightarrow -2} f(x) \text{ does not exist because } \lim_{x \rightarrow -2^-} f(x) = -\infty \text{ and } \lim_{x \rightarrow -2^+} f(x) = \infty.$$

Therefore, f is not continuous at $x = -2$

$$\lim_{x \rightarrow 2} f(x) = 4 = f(2)$$

Therefore, f is continuous at $x = 2$

f is not defined at $x = 3$ Therefore, f is not continuous at $x = 3$.

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Part C

At most 1 out of 3 points if mathematical notation for limit is missing or incorrect. Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ limit notation
- ☐ considers one-sided limits
- ☐ answer

Solution:

As x approaches 0 from the left, $f(x)$ approaches 1 from above. Therefore, the limit of $f(x)$ should be evaluated as x approaches 1 from the right, as shown.

$$\lim_{x \rightarrow 0^-} f(f(x)) = \lim_{x \rightarrow 1^+} f(x) = 4$$

As x approaches 0 from the right, $f(x)$ approaches 2 from above. Therefore, the limit of $f(x)$ should be evaluated as x approaches 2 from the right, as shown.

$$\lim_{x \rightarrow 0^+} f(f(x)) = \lim_{x \rightarrow 2^+} f(x) = 4$$

Therefore, $\lim_{x \rightarrow 0} f(f(x)) = 4$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes a correct difference quotient.

Solution:

$$\frac{g(1.001) - g(0.999)}{1.001 - 0.999} = \frac{5.004 - 4.996}{1.001 - 0.999}$$

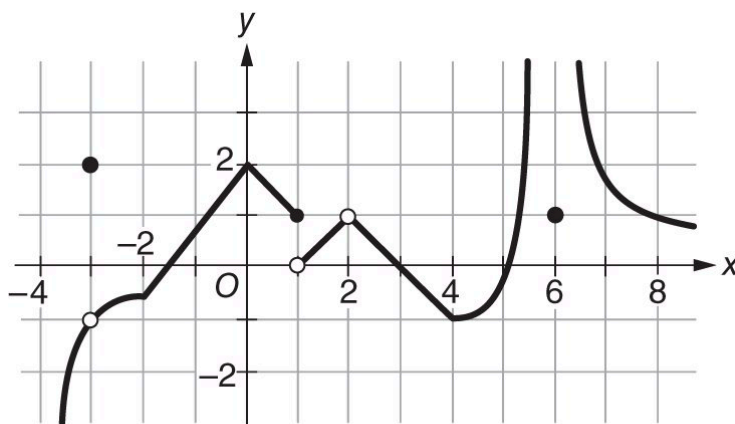
Unit 1 AP Topics 9-12

64. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

x	-0.2	-0.1	-0.01	-0.001	0.001	0.01	0.1	0.2
$g(x)$	1.975	1.988	1.997	1.999	2.001	2.003	2.012	2.025

The graph of the function f is shown in the xy -plane above. f has a discontinuity due to a vertical asymptote at $x = 6$.

Let g be a function that is continuous and increasing for all x . Values of $g(x)$ at selected values of x are shown in the table above.

- Using the graph of f and the table for g , estimate $\lim_{x \rightarrow 0} (f(x) - g(x))$.
- For each of the values $a = 1$, $a = 2$, $a = 4$, and $a = 6$, find $\lim_{x \rightarrow a} f(x)$ or explain why $\lim_{x \rightarrow a} f(x)$ does not exist. Use correct limit notation in your answers.
- Is f continuous at $x = -3$? Using correct notation, justify your answer.
- Write a difference quotient that best approximates the instantaneous rate of change of g at $x = 0$.

Part A

Both limits must be correct to earn the answer point. The response may still earn the points with use of equals rather than approximately.

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Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ estimate for $\lim_{x \rightarrow 0} g(x)$
- ☐ answer

Solution:

$$\lim_{x \rightarrow 0} f(x) = 2 \text{ and } \lim_{x \rightarrow 0} g(x) \approx 2$$

$$\lim_{x \rightarrow 0} (f(x) - g(x)) = \lim_{x \rightarrow 0} f(x) - \lim_{x \rightarrow 0} g(x) \approx 2 - 2 = 0$$

Part B

At most 2 out of 3 points if mathematical notation for any limit is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ does not exist at $a = 1$ with reason
- ☐ values at $a = 2$ and $a = 4$
- ☐ does not exist at $a = 6$ with reason

Solution:

$$\lim_{x \rightarrow 1^-} f(x) = 1 \text{ and } \lim_{x \rightarrow 1^+} f(x) = 0$$

$$\lim_{x \rightarrow 1} f(x) \text{ does not exist because}$$

$$\lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x).$$

$$\lim_{x \rightarrow 2} f(x) = 1$$

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$$\lim_{x \rightarrow 4} f(x) = -1$$

$\lim_{x \rightarrow 6} f(x)$ does not exist because

$$\lim_{x \rightarrow 6^-} f(x) = \infty \text{ and } \lim_{x \rightarrow 6^+} f(x) = \infty.$$

Part C

At most 1 out of 3 points if mathematical notation for limit is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ limit notation
- ☐ limit value
- ☐ answer with justification

Solution:

$$\lim_{x \rightarrow -3} f(x) = -1 \text{ and } f(-3) = 2$$

No, f is not continuous at $x = -3$ because $\lim_{x \rightarrow -3} f(x) \neq f(-3)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes a correct difference quotient.

Solution:

$$\frac{g(0.001) - g(-0.001)}{0.001 - (-0.001)} = \frac{2.001 - 1.999}{0.001 - (-0.001)}$$

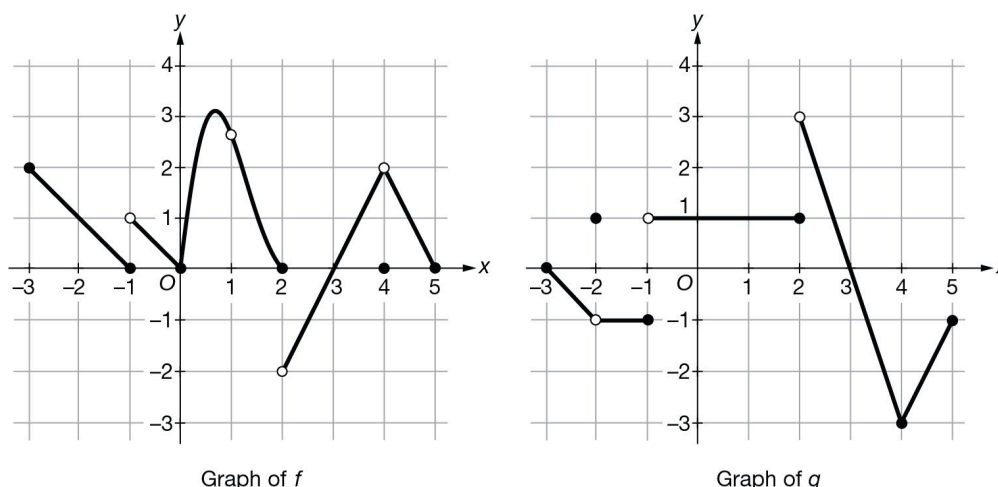
Unit 1 AP Topics 9-12

65. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The graphs of the functions f and g are shown above on the interval $-3 \leq x \leq 5$. The graphs consist of line segments, except where $f(x) = \frac{5 \sin(\pi x)}{2 - 2x^3}$ on the interval $0 \leq x \leq 2$.

- Write a difference quotient that best approximates the instantaneous rate of change of g at $x = 4.5$.
- Let $h(x) = g(f(x))$. Find $\lim_{x \rightarrow 4} h(x)$. Use correct limit notation in your answer.
- Let $k(x) = f(x) + g(x)$. Consider $x = -1$, $x = 2$, and $x = 4$. At which one of these values does k have a jump discontinuity? Justify your answer for the jump discontinuity using correct limit notation.
- Values of $f(x)$ at selected values of x are given in the table below. What type of discontinuity does f have at $x = 1$? Based on the values in the table, what is a reasonable estimate for $\lim_{x \rightarrow 1} f(x)$? Give a reason for your answer.

x	0.995	0.996	0.997	0.998	0.999	1	1.001	1.002	1.003	1.004	1.005
$f(x)$	2.631	2.628	2.626	2.623	2.621	Undefined	2.615	2.613	2.610	2.607	2.605

Part A

Select a point value to view scoring criteria, solutions, **and/or** examples to score the response.

Unit 1 AP Topics 9-12



0	1
---	---

The student response accurately includes a correct difference quotient.

Solution:

$$\frac{g(5)-g(4)}{5-4} = \frac{(-1)-(-3)}{5-4}$$

Part B

At most 2 out of 3 points if mathematical notation for any limit is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ limit notation
- ☐ considers one-sided limits
- ☐ answer

Solution:

As x approaches 4 from the left and from the right, $f(x)$ approaches 2 from below. Therefore, the limit of $g(x)$ should be evaluated as x approaches 2 from the left, as shown.

$$\lim_{x \rightarrow 4^-} h(x) = \lim_{x \rightarrow 4^-} g(f(x)) = \lim_{x \rightarrow 2^-} g(x) = 1$$

$$\lim_{x \rightarrow 4^+} h(x) = \lim_{x \rightarrow 4^+} g(f(x)) = \lim_{x \rightarrow 2^-} g(x) = 1$$

Therefore, $\lim_{x \rightarrow 4} h(x) = 1$.

Part C

At most 1 out of 2 points if mathematical notation for any limit is missing or incorrect OR if response includes jump discontinuity at maximum of one incorrect value.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Unit 1 AP Topics 9-12



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ jump discontinuity at $x = -1$
- ☐ justification, using correct limit notation

Solution:

There is a jump discontinuity at $x = -1$ because both one-sided limits exist at

$x = -1$ and are finite but are not equal:

$$\lim_{x \rightarrow -1^-} k(x) = \lim_{x \rightarrow -1^-} f(x) + \lim_{x \rightarrow -1^-} g(x) = 0 + (-1) = -1$$

$$\lim_{x \rightarrow -1^+} k(x) = \lim_{x \rightarrow -1^+} f(x) + \lim_{x \rightarrow -1^+} g(x) = 1 + 1 = 2$$

Note: k does not have a jump discontinuity at $x = 2$. $\lim_{x \rightarrow 2^-} k(x) = 0 + 1 = 1$ and

$$\lim_{x \rightarrow 2^+} k(x) = -2 + 3 = 1$$

Note: k does not have a jump discontinuity at $x = 4$. $\lim_{x \rightarrow 4^-} k(x) = 2 + (-3) = -1$

$$\text{and } \lim_{x \rightarrow 4^+} k(x) = 2 + (-3) = -1$$

Part D

Any value between 2.615 and 2.621 earns the estimate point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ identifies removable discontinuity
- ☐ estimate
- ☐ reason

Unit 1 AP Topics 9-12**Solution:**

The graph of f indicates that f has a removable discontinuity at $x = 1$.

The graph shows that f is decreasing near $x = 1$, so $2.621 > \lim_{x \rightarrow 1} f(x) > 2.615$. Since the table of values reflects approximate linear behavior, a reasonable estimate for $\lim_{x \rightarrow 1} f(x)$ would be determined by averaging the function values on each side of 1.

The estimate is as follows. $\lim_{x \rightarrow 1} f(x) \approx \frac{f(0.999) + f(1.001)}{2} = \frac{2.621 + 2.615}{2} = 2.618$

Alternately,

a reasonable estimate can be found by observing that as x increases by 0.001, $f(x)$ decreases by 0.002 or 0.003. Since 2.618 is 0.003 from both 2.621 and 2.615, it is the best estimate.

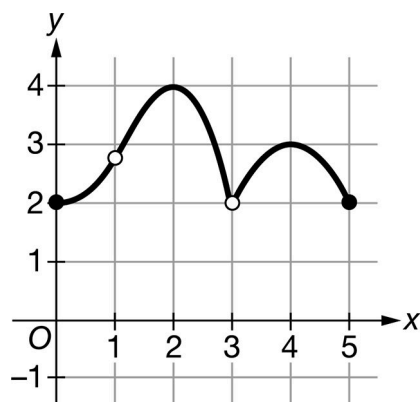
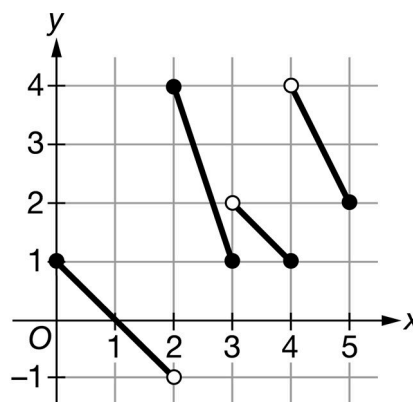
Unit 1 AP Topics 9-12

66. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f Graph of g

The graphs of the functions f and g are shown above on the interval $0 \leq x \leq 5$.

- Write a difference quotient that best approximates the instantaneous rate of change of g at $x = 2.5$.
- Let h be the function defined by $h(x) = g(f(x))$. Find the value of $\lim_{x \rightarrow 3} h(x)$ or explain why the limit does not exist. Use correct limit notation in your answer.
- Let k be the function defined by $k(x) = (4 - f(x))g(x)$. Consider $x = 2$ and $x = 4$. Determine whether k is continuous at each of these values. Justify your answers using correct limit notation.
- Values of $f(x)$ at selected values of x are given in the table below. What type of discontinuity does f have at $x = 1$? Based on the values in the table, what is a reasonable estimate for $\lim_{x \rightarrow 1} f(x)$? Give a reason for your answer.

x	0.9	0.99	0.999	1	1.001	1.01	1.1
$f(x)$	2.634	2.770	2.784	Undefined	2.787	2.801	2.946

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Unit 1 AP Topics 9-12



0	1
---	---

The student response accurately includes a correct difference quotient.

Solution:

$$\frac{g(3)-g(2)}{3-2} = \frac{1-4}{3-2}$$

Part B

At most 2 out of 3 points if mathematical notation for any limit is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ limit notation
- ☐ considers one-sided limits
- ☐ answer

Solution:

As x approaches 3 from the left and from the right, $f(x)$ approaches 2 from above. Therefore, the limit of $g(x)$ should be evaluated as x approaches 2 from the right, as shown.

$$\lim_{x \rightarrow 3^-} h(x) = \lim_{x \rightarrow 3^-} g(f(x)) = \lim_{x \rightarrow 2^+} g(x) = 4$$

$$\lim_{x \rightarrow 3^+} h(x) = \lim_{x \rightarrow 3^+} g(f(x)) = \lim_{x \rightarrow 2^+} g(x) = 4$$

Therefore, $\lim_{x \rightarrow 3} h(x) = 4$.

Part C

At most 1 out of 2 points if mathematical notation for any limit is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Unit 1 AP Topics 9-12



0	1	2
---	---	---

The student response accurately includes two of the criteria below.

- ☐ continuous at $x = 2$ with justification
- ☐ not continuous at $x = 4$ with justification

Solution:

$$\lim_{x \rightarrow 2^-} k(x) = \lim_{x \rightarrow 2^-} (4 - f(x))g(x) = \left(4 - \lim_{x \rightarrow 2^-} f(x)\right) \lim_{x \rightarrow 2^-} g(x) = (4 - 4)(-1) = 0$$

$$\lim_{x \rightarrow 2^+} k(x) = \lim_{x \rightarrow 2^+} (4 - f(x))g(x) = \left(4 - \lim_{x \rightarrow 2^+} f(x)\right) \lim_{x \rightarrow 2^+} g(x) = (4 - 4)(4) = 0$$

$$k(2) = (4 - f(2))g(2) = (4 - 4)(4) = 0$$

Because $\lim_{x \rightarrow 2} k(x) = 0 = k(2)$, k is continuous at $x = 2$.

$$\lim_{x \rightarrow 4^-} k(x) = \lim_{x \rightarrow 4^-} (4 - f(x))g(x) = \left(4 - \lim_{x \rightarrow 4^-} f(x)\right) \lim_{x \rightarrow 4^-} g(x) = (4 - 3)(1) = 1$$

$$\lim_{x \rightarrow 4^+} k(x) = \lim_{x \rightarrow 4^+} (4 - f(x))g(x) = \left(4 - \lim_{x \rightarrow 4^+} f(x)\right) \lim_{x \rightarrow 4^+} g(x) = (4 - 3)(4) = 4$$

Because $\lim_{x \rightarrow 4^-} k(x) \neq \lim_{x \rightarrow 4^+} k(x)$,

$\lim_{x \rightarrow 4} k(x)$ does not exist and therefore k is not continuous at $x = 4$.

Part D

Any value between 2.784 and 2.787 earns the estimate point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ identifies removable discontinuity
- ☐ estimate

Unit 1 AP Topics 9-12

☐☐

reason

Solution:

The graph f of indicates that f has a removable discontinuity at $x = 1$. The graph shows that f is increasing near $x = 1$, so $2.784 < \lim_{x \rightarrow 1} f(x) < 2.787$. A reasonable estimate for $\lim_{x \rightarrow 1} f(x)$ would be determined by averaging the function values on each side of 1.

The estimate is as follows.

$$\lim_{x \rightarrow 1} f(x) \approx \frac{f(0.999) + f(1.001)}{2} = \frac{2.784 + 2.787}{2} = 2.7855$$

Unit 1 AP Topics 9-12

67. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$N(t) = \begin{cases} f(t) & \text{for } 0 \leq t < 6 \\ 25t + 150 & \text{for } 6 \leq t < 8 \\ \frac{200+80t}{2+0.05t} & \text{for } t \geq 8 \end{cases}$$

The number of fish in a pond at time t years is modeled by the function N defined above, where f is a continuous function such that $f(0) = 80$.

- (a) Find $\lim_{t \rightarrow \infty} N(t)$. Explain the meaning of $\lim_{t \rightarrow \infty} N(t)$ in the context of the problem.
- (b) Is the function N continuous at $t = 8$? Justify your answer.
- (c) The function N is continuous at $t = 6$. Is there a time t , for $0 \leq t \leq 6$, at which $N(t) = 250$? Justify your answer.

Part A

The second point requires a reference to time increasing and the connection to the number of fish.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\lim_{t \rightarrow \infty} N(t)$
- ☐ interpretation

Solution:

Unit 1 AP Topics 9-12

$$\lim_{t \rightarrow \infty} N(t) = \lim_{t \rightarrow \infty} \frac{200+80t}{2+0.05t} = \frac{80}{0.05} = 1600$$

As time increases, the number of fish in the pond approaches 1600.

Part B

At most 2 out of 3 points if mathematical notation for limits is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\lim_{t \rightarrow 8^-} N(t)$
- ☐ $\lim_{t \rightarrow 8^+} N(t)$
- ☐ answer with justification

Solution:

$$\lim_{t \rightarrow 8^-} N(t) = \lim_{t \rightarrow 8^-} (25t + 150) = 350$$

$$\lim_{t \rightarrow 8^+} N(t) = \lim_{t \rightarrow 8^+} \frac{200+80t}{2+0.05t} = \frac{840}{2.4} = 350$$

$$N(8) = 350$$

Yes, N is continuous at $t = 8$ because

$$\lim_{t \rightarrow 8} N(t) = N(8).$$

Part C

At most 1 out of 2 points if correct justification does not specifically reference IVT

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes all of the criteria below.

Unit 1 AP Topics 9-12

- ☐ answer with justification
- ☐ specific reference to IVT

Solution:

N is continuous for $0 \leq t \leq 6$.

$$N(0) = 80 < 250 < 300 = N(6)$$

By the Intermediate Value Theorem, there must be at least one time t , for $0 \leq t \leq 6$, such that

$$N(t) = 250.$$

Unit 1 AP Topics 9-12

68. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$F(t) = \begin{cases} 2^{t+1} - 2 & \text{for } 0 \leq t < 7 \\ \frac{800t - 4800}{t - 4} & \text{for } t \geq 7 \end{cases}$$

The amount of money raised during a fund-raising campaign is modeled by the function F defined above, where $F(t)$ is measured in United States dollars and t is the time in days since the campaign began.

- Find $\lim_{t \rightarrow \infty} F(t)$. Explain the meaning of $\lim_{t \rightarrow \infty} F(t)$ in the context of the problem.
- Is the function F continuous at $t = 7$? Justify your answer.
- The amount of money raised during a competing fund-raising campaign is modeled by the function M defined by $M(t) = \frac{240(2^t - 1)}{(2^t + 36)}$, where $M(t)$ is measured in United States dollars and t is the time in days since that campaign began. According to this model, is there a time t , for $0 \leq t \leq 2$, at which the amount of money raised is 10 dollars? Justify your answer.

Part A

The second point requires a reference to time increasing and the connection to the amount raised.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\lim_{t \rightarrow \infty} F(t)$
- ☐ interpretation

Solution:

Unit 1 AP Topics 9-12

$$\lim_{t \rightarrow \infty} F(t) = \lim_{t \rightarrow \infty} \frac{800t - 4800}{t - 4} = \frac{800}{1} = 800$$

As time increases, the amount raised by the fund-raising campaign approaches \$800.

Part B

At most 2 out of 3 points if mathematical notation for limits is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\lim_{t \rightarrow 7^-} F(t)$
- ☐ $\lim_{t \rightarrow 8^+} F(t)$
- ☐ answer with justification

Solution:

$$\lim_{t \rightarrow 7^-} F(t) = \lim_{t \rightarrow 7^-} (2^{t+1} - 2) = 2^8 - 2 = 254$$

$$\lim_{t \rightarrow 7^+} F(t) = \lim_{t \rightarrow 7^+} \frac{800t - 4800}{t - 4} = \frac{800 \cdot 7 - 4800}{7 - 4} = \frac{800}{3}$$

No, F is not continuous at $t = 7$ because $\lim_{t \rightarrow 7^-} F(t) \neq \lim_{t \rightarrow 7^+} F(t)$.

Part C

At most 1 out of 2 points if correct justification does not specifically reference IVT

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes all of the criteria below.

- ☐ answer with justification
- ☐ specific reference to IVT

Unit 1 AP Topics 9-12**Solution:**

M is continuous for $0 \leq t \leq 2$.

$$M(0) = \frac{240(2^0 - 1)}{2^0 + 36} = \frac{240(1 - 1)}{1 + 36} = 0$$

$$M(2) = \frac{240(2^2 - 1)}{2^2 + 36} = \frac{240 \cdot 3}{40} = 18$$

$$M(0) = 0 < 10 < 18 = M(2)$$

By the Intermediate Value Theorem, there must be at least one time t , for $0 \leq t \leq 2$, such that $M(t) = 10$.

Unit 1 AP Topics 9-12

69. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$R(t) = \begin{cases} 41\sqrt{\frac{t}{20}} & \text{for } 0 \leq t < 20 \\ \frac{100t^2 + 1000}{t^2 + 600} & \text{for } t \geq 20 \end{cases}$$

The total accumulated revenue a company has received up to time t is modeled by the function R defined above, where $R(t)$ is measured in millions of United States dollars and t is the time in years since 1980.

- Find $\lim_{t \rightarrow \infty} R(t)$. Explain the meaning of $\lim_{t \rightarrow \infty} R(t)$ in the context of the problem.
- Is the function R continuous at $t = 20$? Justify your answer.
- The company's total accumulated expenses up to time t is modeled by the function E defined by $E(t) = 3\log_2(12t + 4)$, where $E(t)$ is measured in millions of United States dollars and t is the time in years since 1980. According to these models, is there a time t , for $0 \leq t \leq 5$, at which the total accumulated profit, $R(t) - E(t)$, is equal to 0? Justify your answer.

Part A

The second point requires a reference to time increasing and the connection to the revenue of the company.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\lim_{t \rightarrow \infty} R(t)$
- ☐ interpretation

Solution:

Unit 1 AP Topics 9-12

$$\lim_{t \rightarrow \infty} R(t) = \lim_{t \rightarrow \infty} \frac{100t^2 + 1000}{t^2 + 600} = \frac{100}{1} = 100$$

As time increases, the total accumulated revenue of the company approaches \$100 million.

Part B

At most 2 out of 3 points if mathematical notation for limits is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\lim_{t \rightarrow 20^-} R(t)$
- ☐ $\lim_{t \rightarrow 20^+} R(t)$
- ☐ answer with justification

Solution:

$$\lim_{t \rightarrow 20^-} R(t) = \lim_{t \rightarrow 20^-} 41\sqrt{\frac{t}{20}} = 41\sqrt{\frac{20}{20}} = 41$$

$$\lim_{t \rightarrow 20^+} R(t) = \lim_{t \rightarrow 20^+} \frac{100t^2 + 1000}{t^2 + 600} = \lim_{t \rightarrow 20^+} \frac{100(t^2 + 10)}{t^2 + 600} = \frac{100(20^2 + 10)}{20^2 + 600} = \frac{100(410)}{1000} = 41$$

$$R(20) = 41$$

Yes, R is continuous at $t = 20$ because $\lim_{t \rightarrow 20} R(t) = R(20)$.

Part C

At most 1 out of 2 points if correct justification does not specifically reference IVT.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes all of the criteria below.

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- ☐ answer with justification
- ☐ specific reference to IVT.

Solution:

R and E are continuous for $0 \leq t \leq 5$.

Therefore, the function defined by $R(t) - E(t)$ is continuous for $0 \leq t \leq 5$.

$$R(0) - E(0) = 41\sqrt{\frac{0}{20}} - 3\log_2(12 \cdot 0 + 4) = 0 - 6 = -6 < 0$$

$$\begin{aligned} R(5) - E(5) &= 41\sqrt{\frac{5}{20}} - 3\log_2(12 \cdot 5 + 4) \\ &= \frac{41}{2} - 3\log_2 64 = 20.5 - 18 > 0 \end{aligned}$$

$$R(0) - E(0) < 0 < R(5) - E(5)$$

By the Intermediate Value Theorem, there must be at least one time t , for $0 \leq t \leq 5$, such that $R(t) - E(t) = 0$.

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70. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$Y(t) = \begin{cases} 10e^{0.05t} & \text{for } 0 \leq t \leq 10 \\ f(t) & \text{for } 10 < t \leq 12 \\ \frac{600}{20+10e^{-0.05(t-12)}} & \text{for } t > 12 \end{cases}$$

The mass of yeast in a culture, measured in milligrams, at time t hours is modeled by the function Y defined above, where f is a continuous function such that $f(12) = 20$.

Find $\lim_{t \rightarrow \infty} Y(t)$. Explain the meaning of $\lim_{t \rightarrow \infty} Y(t)$ in the context of the problem.

Is the function Y continuous at $t = 12$? Justify your answer.

The function Y is continuous at $t = 10$. Is there a time t , for $0 < t < 12$, at which $Y(t) = 18$? Justify your answer.

Part A

The second point requires a reference to time increasing and the connection to the mass of the yeast in the culture.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\lim_{t \rightarrow \infty} Y(t)$
- ☐ interpretation

Solution:

$$\lim_{t \rightarrow \infty} Y(t) = \lim_{t \rightarrow \infty} \frac{600}{20+10e^{-0.05(t-12)}} = \frac{600}{20+0} = 30$$

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As time increases, the mass of the yeast in the culture approaches 30 milligrams.

Part B

At most 2 out of 3 points if mathematical notation for limits is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\lim_{t \rightarrow 12^-} Y(t)$
- ☐ $\lim_{t \rightarrow 12^+} Y(t)$
- ☐ answer with justification

Solution

Because f is a continuous function and $f(12) = 20$, $\lim_{t \rightarrow 12^-} Y(t) = \lim_{t \rightarrow 12^-} f(t) = f(12) = 20$.

$$\lim_{t \rightarrow 12^+} Y(t) = \lim_{t \rightarrow 12^+} \frac{600}{20 + 10e^{-0.05(t-12)}} = \frac{600}{20 + 10} = 20$$

$$Y(12) = f(12) = 20$$

Yes, Y is continuous at $t = 12$ because $\lim_{t \rightarrow 12} Y(t) = Y(12)$.

Part C

At most 1 out of 2 points if correct justification does not specifically reference IVT.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes all of the criteria below.

- ☐ answer with justification
- ☐ specific reference to *IVT*

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Solution:

Y is continuous for $0 \leq t \leq 12$.

$$Y(0) = 10 < 18 < 20 = Y(12)$$

By the Intermediate Value Theorem, there must be at least one time t , $0 \leq t \leq 12$, such that $Y(t) = 18$.

71. Which of the following functions is not continuous on the interval $-\infty < x < \infty$?

(A) $f(x) = x^4 + x^3 + x^2 + x + 1$

(B) $g(x) = \frac{1}{x^3 + x^2 + x + 1}$

(C) $h(x) = \frac{\pi}{2} \sin x$

(D) $k(x) = \frac{1}{1 + e^{-x}}$



Answer B

Correct. A rational function is continuous on all points in its domain. Since a cubic polynomial must have at least one zero, there is at least one value of x that is not in the domain of the rational function g . Because there is at least one x in the interval $-\infty < x < \infty$ where g is not continuous, the function is not continuous on the interval $-\infty < x < \infty$.

72. Which of the following functions are continuous on the interval $0 < x < 2$?

I. $f(x) = \frac{x-1}{x^2-1}$

II. $g(x) = \frac{x+1}{x^2+1}$

III. $h(x) = \ln(x^2 - 1)$

(A) II only

(B) I and II only

(C) I and III only

(D) II and III only



Answer A

Correct. A rational function is continuous at all points in its domain. The domain of the rational function g is all real numbers x , since the denominator of g is always nonzero. Therefore, g is continuous on any interval.

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73.  Let f be the function given by $f(x) = \frac{|x^2-3| \cdot (x+0.5)}{(x^2-3)(x+0.5)}$. On which of the following open intervals is f continuous?

(A) $(-2, -1)$

(B) $(-1, 0)$

(C) $(0, 1)$

(D) $(1, 2)$

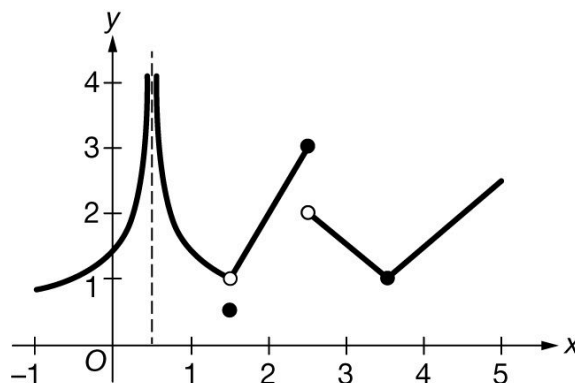
**Answer C**

Correct. Graphing f on the calculator visually reveals two jump discontinuities on the intervals $(-2, -1)$ and $(1, 2)$. However, f is discontinuous everywhere the denominator equals zero.

$(x^2 - 3)(x + 0.5) = 0 \Rightarrow x \approx -1.732, x = -0.5, x \approx 1.732$ In addition to the two jump discontinuities, there is a removable discontinuity at $x = -0.5$. Therefore, the function is continuous for $0 < x < 1$, since none of the zeros occur in that interval.

Unit 1 AP Topics 9-12

74.

Graph of f

The graph of the function f is shown above. On which of the following intervals is f continuous?

- (A) $(-1, 1)$
(B) $(1, 2)$
(C) $(2, 3)$
(D) $(3, 5)$

**Answer D**

Correct. The graph indicates that f is continuous at each point in the interval $(3, 5)$, and therefore f is continuous on that interval.

75. The function f is continuous on the interval $-1 < x < 3$ and is not continuous on the interval $-1 \leq x \leq 3$. Which of the following could not be an expression for $f(x)$?

- (A) $\frac{x+1}{x-3}$
(B) $\frac{x-3}{x+1}$
(C) $(x+1)(x-3)$
(D) $\frac{1}{(x+1)(x-3)}$

**Answer C**

Correct. A polynomial is continuous on all points in its domain. Since the function f is not continuous on the interval $-1 \leq x \leq 3$, it cannot be a polynomial.

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76.

$$f(x) = \begin{cases} \sin x & \text{for } x < 0 \\ \cos x & \text{for } 0 \leq x \leq \frac{3\pi}{2} \\ \tan x & \text{for } \frac{3\pi}{2} < x \leq 2\pi \\ \cot x & \text{for } 2\pi < x \leq \frac{5\pi}{2} \end{cases}$$

Let f be the function given above. On which of the following intervals is f continuous?

(A) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(B) $\left(\frac{\pi}{4}, \pi\right)$

(C) $\left(\pi, \frac{7\pi}{4}\right)$

(D) $\left(\frac{7\pi}{4}, \frac{5\pi}{2}\right)$

**Answer B**

Correct. The function $\cos x$ is continuous for all x , and $f(x) = \cos x$ on an interval that includes $\left(\frac{\pi}{4}, \pi\right)$. Therefore, for all x in the interval $\left(\frac{\pi}{4}, \pi\right)$, $f(x)$ is continuous.

77.

$$f(x) = \begin{cases} e^x & \text{for } x \leq -1 \\ x^3 + 2x & \text{for } -1 < x \leq 0 \\ \frac{5x}{x-2} & \text{for } 0 < x \leq 4 \\ \cos(3x) & \text{for } x > 4 \end{cases}$$

Let f be the function given above. On which of the following intervals is f continuous?

(A) $(-5, 0)$

(B) $(-1, 2)$

(C) $(0, 3)$

(D) $(2, 5)$

**Answer B**

Correct. The function f is continuous at $x = 0$ since $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (x^3 + 2x) = 0$,

$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{5x}{x-2} = 0$, and $f(0) = 0$. The polynomial $x^3 + 2x$ is continuous on the interval $(-1, 0)$, and the rational function $\frac{5x}{x-2}$ is continuous on the interval $(0, 2)$. Therefore, f is continuous on $(-1, 2)$.

Unit 1 AP Topics 9-12

78. Which of the following functions is not continuous on the interval $-\infty < x < \infty$?

(A) $f(x) = 4x^2 - 2x + 1$

(B) $g(x) = \frac{1}{x^3 + 3x^2 - 2x - 5}$

(C) $h(x) = \cos(\pi x)$

(D) $k(x) = \frac{1}{e^x}$

**Answer B**

Correct. A rational function is continuous on all points in its domain. Since a cubic polynomial must have at least one zero, there is at least one value of x that is not in the domain of the rational function g . Because there is at least one x in the interval $-\infty < x < \infty$ where g is not continuous, the function is not continuous on the interval $-\infty < x < \infty$.

79. Which of the following functions are continuous on the interval $0 < x < 5$?

I. $f(x) = \frac{x-3}{x^2-9}$

II. $g(x) = \frac{x-3}{x^2+9}$

III. $h(x) = \ln(x-3)$

(A) II only

(B) I and II only

(C) I and III only

(D) II and III only

**Answer A**

Correct. The domain of g is all x , and therefore g is continuous for all x .

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80.  Let f be the function given by $f(x) = \frac{|x^2-2|(x+0.4)}{(x^2-2)(x+0.4)}$. On which of the following open intervals is f continuous?

(A) $(-2, -1)$

(B) $(-1, 0)$

(C) $(0, 1)$

(D) $(1, 2)$

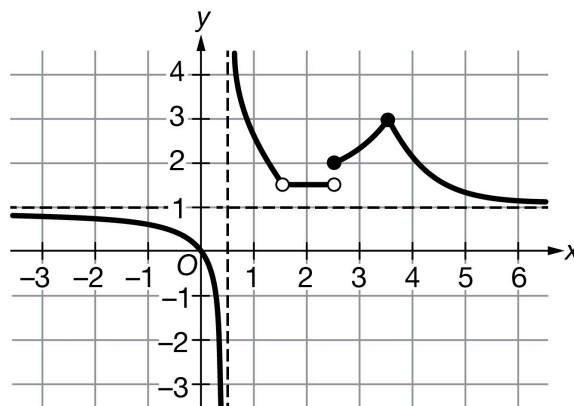
**Answer C**

Correct. Graphing f on the calculator visually reveals two jump discontinuities on the intervals $(-2, -1)$ and $(1, 2)$. Moreover, f is discontinuous everywhere the denominator equals zero.

$(x^2 - 2)(x + 0.4) = 0 \Rightarrow x \approx -1.414, x = -0.4, x \approx 1.414$. In addition to the two jump discontinuities, there is a removable discontinuity at $x = -0.4$. Therefore, the function is continuous for $0 < x < 1$, since none of the zeros of the denominator occur in that interval.

Unit 1 AP Topics 9-12

81.

Graph of f

The graph of the function f is shown above. On which of the following intervals is f continuous?

- (A) $(0, 1)$
(B) $(1, 2)$
(C) $(2, 3)$
(D) $(3, 4)$

**Answer D**

Correct. The graph indicates that f is continuous at each point in the interval $(3, 4)$, and therefore f is continuous on that interval.

82. The function f is continuous on the interval $-2 < x < 5$ and is not continuous on the interval $-2 \leq x \leq 5$. Which of the following could not be an expression for $f(x)$?

- (A) $\frac{x+2}{x-5}$
(B) $\frac{x-5}{x+2}$
(C) $(x+2)(x-5)$
(D) $\frac{1}{(x+2)(x-5)}$

**Answer C**

Correct. A polynomial is continuous on all points in its domain. Since the function f is not continuous on the interval $-2 \leq x \leq 5$, it cannot be a polynomial.

Unit 1 AP Topics 9-12

83.  A 12,000-liter tank of water is filled to capacity. At time $t = 0$, water begins to drain out of the tank at a rate modeled by $r(t)$, measured in liters per hour, where r is given by the piecewise-defined function

$$r(t) = \begin{cases} \frac{600t}{t+3} & \text{for } 0 \leq t \leq 5 \\ 1000e^{-0.2t} & \text{for } t > 5 \end{cases}$$

- (a) Is r continuous at $t = 5$? Show the work that leads to your answer.
 (b) Find the average rate at which water is draining from the tank between time $t = 0$ and time $t = 8$ hours.
 (c) Find $r'(3)$. Using correct units, explain the meaning of that value in the context of this problem.
 (d) Write, but do not solve, an equation involving an integral to find the time A when the amount of water in the tank is 9000 liters.

Part A

2 points are earned for conclusion with analysis

$$\lim_{t \rightarrow 5^-} r(t) = \lim_{t \rightarrow 5^-} \left(\frac{600t}{t+3} \right) = 375 = r(5)$$

$$\lim_{t \rightarrow 5^+} r(t) = \lim_{t \rightarrow 5^+} (1000e^{-0.2t}) = 367.879$$

Because the left-hand and right-hand limits are not equal, r is not continuous at $t = 5$.



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for conclusion with analysis

$$\lim_{t \rightarrow 5^-} r(t) = \lim_{t \rightarrow 5^-} \left(\frac{600t}{t+3} \right) = 375 = r(5)$$

$$\lim_{t \rightarrow 5^+} r(t) = \lim_{t \rightarrow 5^+} (1000e^{-0.2t}) = 367.879$$

Because the left-hand and right-hand limits are not equal, r is not continuous at $t = 5$.

Part B

1 point is earned for the integrand

1 point is earned for the limits and constant

1 point is earned for the answer

Unit 1 AP Topics 9-12

$$\frac{1}{8} \int_0^8 r(t) dt = \frac{1}{8} \left(\int_0^5 \frac{600t}{t+3} dt + \int_5^8 1000e^{-0.2t} dt \right)$$

$$= 258.052 \text{ or } 258.053$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits and constant

1 point is earned for the answer

$$\frac{1}{8} \int_0^8 r(t) dt = \frac{1}{8} \left(\int_0^5 \frac{600t}{t+3} dt + \int_5^8 1000e^{-0.2t} dt \right)$$

$$= 258.052 \text{ or } 258.053$$

Part C

1 point is earned for $r'(3)$

1 point is earned for the meaning of $r'(3)$

$$r'(3) = 50$$

The rate at which water is draining out of the tank at time $t = 3$ hours is increasing at 50 liters/hour².



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $r'(3)$

1 point is earned for the meaning of $r'(3)$

$$r'(3) = 50$$

Unit 1 AP Topics 9-12

The rate at which water is draining out of the tank at time $t = 3$ hours is increasing at 50 liters/hour².

Part D

1 point is earned for the integral

1 point is earned for the equation

$$12,000 - \int_0^A r(t)dt = 9000$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the equation

$$12,000 - \int_0^A r(t)dt = 9000$$

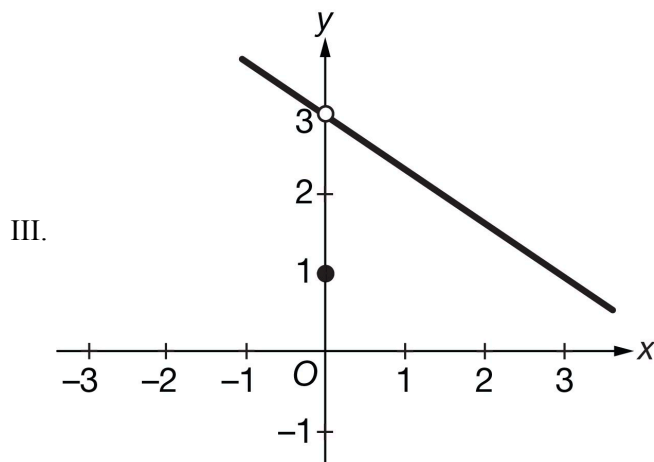
Unit 1 AP Topics 9-12

84. Let g be a function that is decreasing for $x < 0$ and decreasing for $x > 0$. If $\lim_{x \rightarrow 0} g(x) = 3$, which of the following could represent the function g ?

I. $g(x) = \begin{cases} -\frac{x^2-3x}{x} & \text{for } x \neq 0 \\ 5 & \text{for } x = 0 \end{cases}$

II.

x	-0.2	-0.1	-0.01	-0.001	0	0.001	0.01	0.1
$g(x)$	-10	-100	-1000	-10,000	3	10,000	1000	100



- (A) I only
 (B) II only
 (C) III only
 (D) I and III only



Answer D

Correct. Since

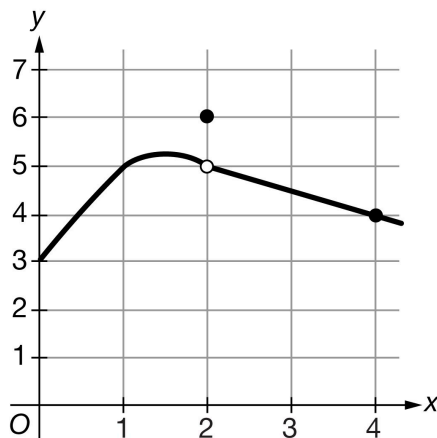
$$\lim_{x \rightarrow 0} -\frac{x^2-3x}{x} = \lim_{x \rightarrow 0} -\frac{x(x-3)}{x} = \lim_{x \rightarrow 0} (-x + 3) = 0 + 3 = 3,$$

the expression in I could represent the function g . The graph in III could also represent the function g , since the limit determined by the graph as x approaches 0 is equal to 3. The table in II could not represent the function g , since the values in the table suggest that the function grows without bound as x approaches 0 from both sides, and thus the limit would not exist at $x = 0$.

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85.

$$f(x) = \begin{cases} -x^2 + 3x + 3 & \text{for } x < 2 \\ 6 & \text{for } x = 2 \\ 6 - \frac{1}{2}x & \text{for } x > 2 \end{cases}$$



Let f be the piecewise function defined above. Also shown is a portion of the graph of f . What is the value of $\lim_{x \rightarrow 2} f(f(x))$?

- (A) -15
 (B) -7
 (C) 3
 (D) $\frac{7}{2}$

**Answer D**

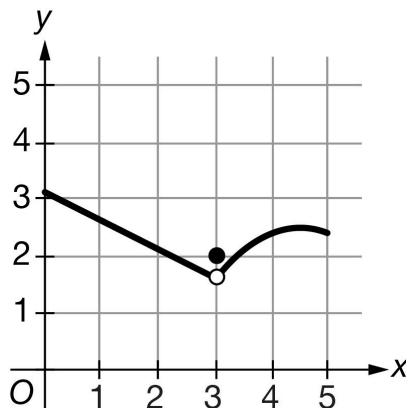
Correct. The graph of f indicates that $\lim_{x \rightarrow 2} f(x) = 5$. (This can be confirmed by noting that

$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (-x^2 + 3x + 3) = -4 + 6 + 3 = 5$ and $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (6 - \frac{1}{2}x) = 6 - 1 = 5$.) The behavior of f near $x = 5$ is not visible in the graph of f , but finding the limit with the function representation shows that $\lim_{x \rightarrow 2} f(f(x)) = \lim_{x \rightarrow 5} f(x) = \lim_{x \rightarrow 5} (6 - \frac{1}{2}x) = 6 - \frac{5}{2} = \frac{7}{2}$.

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86.

x	2.9	2.95	2.99	2.998	3.002	3.01	3.05	3.1
$f(x)$	1.650	1.625	1.605	1.601	1.602	1.612	1.659	1.716



The table above gives selected values for a function f . Also shown is a portion of the graph of f . The graph consists of a line segment for $x < 3$ and part of a parabola for $x > 3$. What is $\lim_{x \rightarrow 3} f(x)$?

- (A) 1.6
- (B) $\frac{1.601+1.602}{2}$
- (C) 2
- (D) The limit does not exist.



Answer A

Correct. The graph of f indicates that the limit exists at $x = 3$ and has a value between 1 and 2. The linear segment in the graph of f has slope $\frac{1.625-1.650}{2.95-2.9} = \frac{-0.025}{0.05} = -\frac{1}{2}$. The equation of the line is $y = -\frac{1}{2}(x - 2.9) + 1.65$. Therefore, $\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} \left(-\frac{1}{2}(x - 2.9) + 1.65\right) = -0.05 + 1.65 = 1.6$.

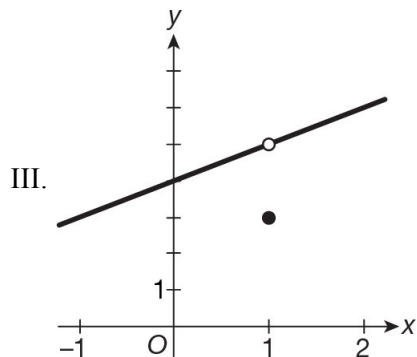
Unit 1 AP Topics 9-12

87. Let g be a function that is increasing for $x < 1$ and increasing for $x > 1$. If $\lim_{x \rightarrow 1} g(x) = 5$, which of the following could represent the function g ?

I. $g(x) = \begin{cases} \frac{x^2+3x-4}{x-1} & \text{for } x \neq 1 \\ 3 & \text{for } x = 1 \end{cases}$

II.

x	0.8	0.9	0.95	0.999	1	1.001	1.05	1.1	1.2
$g(x)$	25	50	100	5000	5	-5000	-100	-50	-25



- (A) I only
 (B) II only
 (C) III only
 (D) I and III only



Answer D

Correct. Since $\lim_{x \rightarrow 1} \frac{x^2+3x-4}{x-1} = \lim_{x \rightarrow 1} \frac{(x-1)(x+4)}{x-1} = \lim_{x \rightarrow 1} (x+4) = 5$, the expression in (I) could represent the function g .

The graph in (III) could also represent the function g , since the limit determined by the graph as x approaches 1 is equal to 5. This is the graph of the function in (I).

The table in (II) could not represent the function g , since the values in the table suggest that the function grows without bound approaching $x = 1$ from both sides, and thus the limit would not exist at $x = 1$.

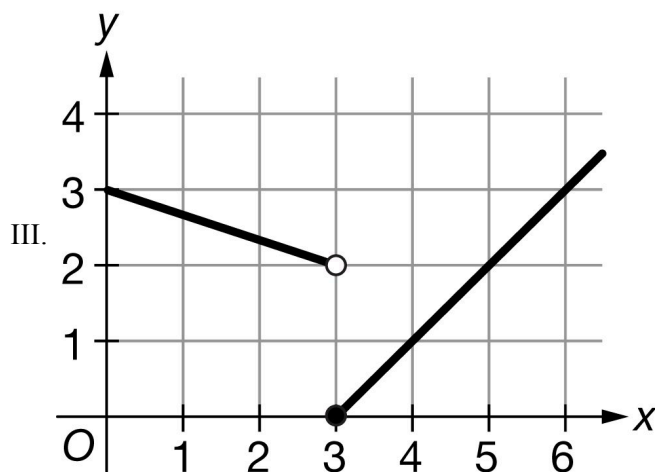
Unit 1 AP Topics 9-12

88. If h is a piecewise linear function such that $\lim_{x \rightarrow 3} h(x)$ does not exist, which of the following could be representative of the function h ?

I. $h(x) = \begin{cases} 12 - x & \text{for } x < 3 \\ 4x - 3 & \text{for } x > 3 \end{cases}$

II.

x	0	1	2	3	4	5	6
$h(x)$	3	$\frac{8}{3}$	$\frac{7}{3}$	0	1	2	3



- (A) I only
 (B) II only
 (C) III only
 (D) II and III only



Answer D

Correct. Given the data in the table for $x < 3$, it appears that $\lim_{x \rightarrow 3^-} h(x) = 2$. The data in the table for $x > 3$ suggest that $\lim_{x \rightarrow 3^+} h(x) = 0$. Since the right-hand and left-hand limits are not equal, the limit at $x = 3$ does not appear to exist for the function with selected values in this table. The graph in III could also represent the function h , since the graph shows a jump discontinuity at $x = 3$, where the left- and right-hand limits are unequal. The values in the table in II are the same as the values on this graph. The expression in I could not represent the function h , since $\lim_{x \rightarrow 3^-} h(x) = 12 - x = 12 - 3 = 9$ and $\lim_{x \rightarrow 3^+} h(x) = 4(3) - 3 = 12 - 3 = 9$, which implies that the limit of the expression in I exists at $x = 3$.

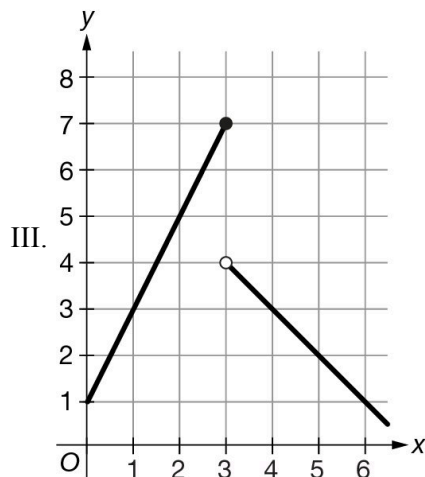
Unit 1 AP Topics 9-12

89. If h is a piecewise linear function such that $\lim_{x \rightarrow 3} h(x)$ does not exist, which of the following could represent the function h ?

I. $h(x) = \begin{cases} 2x + 1 & \text{for } x < 3 \\ \text{undefined} & \text{for } x = 3 \\ 10 - x & \text{for } x > 3 \end{cases}$

II.

x	2.6	2.7	2.8	2.9	3	3.1	3.2	3.3	3.4
$h(x)$	6.2	6.4	6.6	6.8	7	3.9	3.8	3.7	3.6



- (A) I only
 (B) II only
 (C) III only
 (D) II and III only



Answer D

Correct. The table in (II) could represent the function h . The data in the table for $x < 3$ are linear with slope 2, so the limit as x approaches 3 from the left would be 7. The data in the table for $x > 3$ are also linear with slope -1 , so the limit as x approaches 3 from the right would be 4. Since the left- and right-hand limits are not equal, the limit at $x = 3$ would not exist for the function represented by this table.

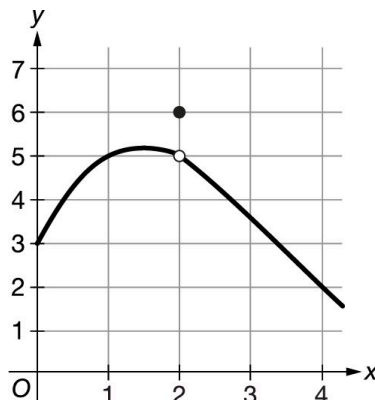
The graph in (III) could also represent the function h , since the graph shows a jump discontinuity at $x = 3$, where the left- and right-hand limits are unequal. The values in the table in (II) are the same as the values on this graph.

The expression in (I) could not represent the function h , since $\lim_{x \rightarrow 3^-} (2x + 1) = 7$ and $\lim_{x \rightarrow 3^+} (10 - x) = 7$ implies that the limit of the expression in (I) exists at $x = 3$.

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90.

$$f(x) = \begin{cases} -x^2 + 3x + 3 & \text{for } x < 2 \\ 6 & \text{for } x = 2 \\ 8 - \frac{3}{2}x & \text{for } x > 2 \end{cases}$$



Let f be the piecewise function defined above. Also shown is a portion of the graph of f . What is the value of $\lim_{x \rightarrow 2} f(f(x))$?

- (A) -15
 (B) -7
 (C) -1
 (D) $\frac{1}{2}$

**Answer D**

Correct. The graph of f indicates that $\lim_{x \rightarrow 2} f(x) = 5$. (This can be confirmed by noting that

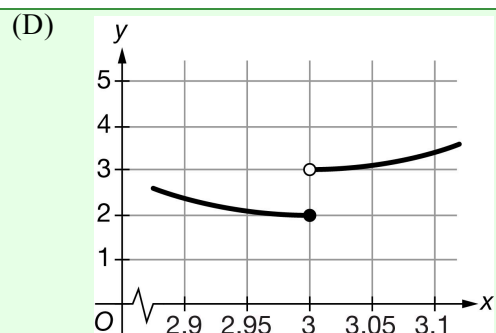
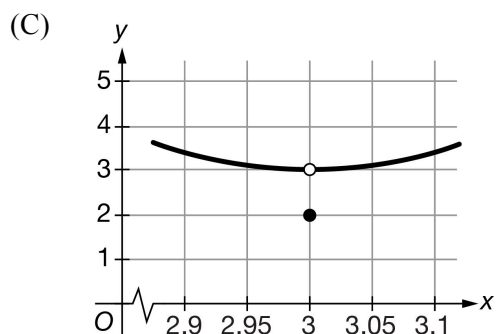
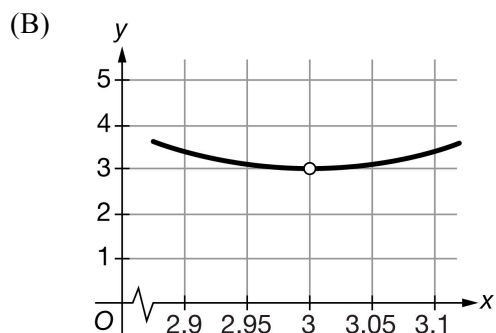
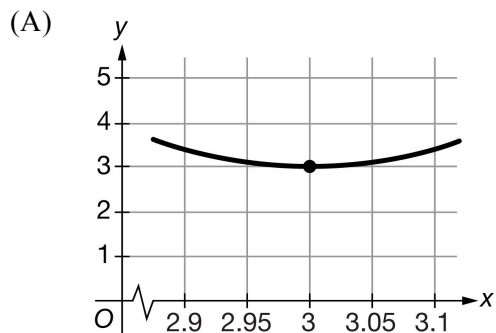
$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (-x^2 + 3x + 3) = -4 + 6 + 3 = 5$ and $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (8 - \frac{3}{2}x) = 8 - 3 = 5$.) The behavior of f near $x = 5$ is not visible in the graph of f , but finding the limit with the function representation shows that $\lim_{x \rightarrow 2} f(f(x)) = \lim_{x \rightarrow 5} f(x) = \lim_{x \rightarrow 5} (8 - \frac{3}{2}x) = 8 - \frac{15}{2} = \frac{1}{2}$.

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91.

x	2.9	2.95	2.99	2.999	3.001	3.01	3.05	3.1
$f(x)$	3.4	3.1	3.004	3.00004	3.00004	3.004	3.1	3.4

The table above gives selected values for a function f . Based on the data in the table, which of the following could not be the graph of f on the interval $2.9 \leq x \leq 3.1$?

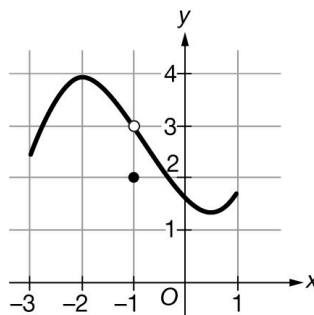


Answer D

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Correct. The data in the table suggest that $\lim_{x \rightarrow 3} f(x) = 3$. This function has a jump discontinuity at $x = 3$ where the limit does not exist. Therefore, this could not be the graph of f .

92.

Graph of f

The graph of the function f is shown above. Which of the following could be a table of values for f ?

(A)

x	-1.05	-1.001	-1	-0.999	-0.95
$f(x)$	3.074	3.001	2	1.998	1.924

(B)

x	-1.05	-1.001	-1	-0.999	-0.95
$f(x)$	3.074	3.001	2	2.998	2.924

(C)

x	-1.05	-1.001	-1	-0.999	-0.95
$f(x)$	3.074	3.001	3	2.998	2.924

(D)

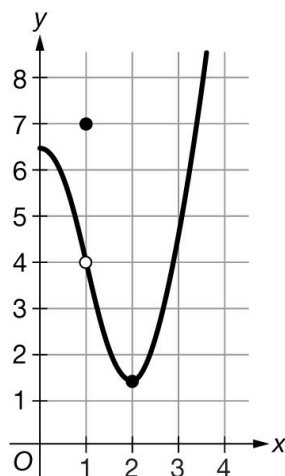
x	-1.05	-1.001	-1	-0.999	-0.95
$f(x)$	3.074	3.001	undefined	2.998	2.924

Answer B

Correct. The graph of f shows that $\lim_{x \rightarrow -1} f(x) = 3$ and $f(-1) = 2$. This table is consistent with that information.

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93.

Graph of f

The graph of the function f is shown above. Which of the following could be a table of values for f ?

(A)

x	0.95	0.99	1	1.001	1.05
$f(x)$	6.9025	6.9801	7	7.0020	7.1025

(B)

x	0.95	0.99	1	1.001	1.05
$f(x)$	4.1624	4.0325	7	3.9968	3.8376

(C)

x	0.95	0.99	1	1.001	1.05
$f(x)$	4.1624	4.0325	4	3.9968	3.8376

(D)

x	0.95	0.99	1	1.001	1.05
$f(x)$	4.1624	4.0325	undefined	3.9968	3.8376

Answer B

Correct. The graph of f shows that $\lim_{x \rightarrow 1} f(x) = 4$ and $f(1) = 7$. This table is consistent with that information.

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94.

x	1.9	1.95	1.99	1.999	2.001	2.01	2.05	2.1
$f(x)$	4.204	3.671	3.147	3.015	2.985	2.847	2.160	1.113

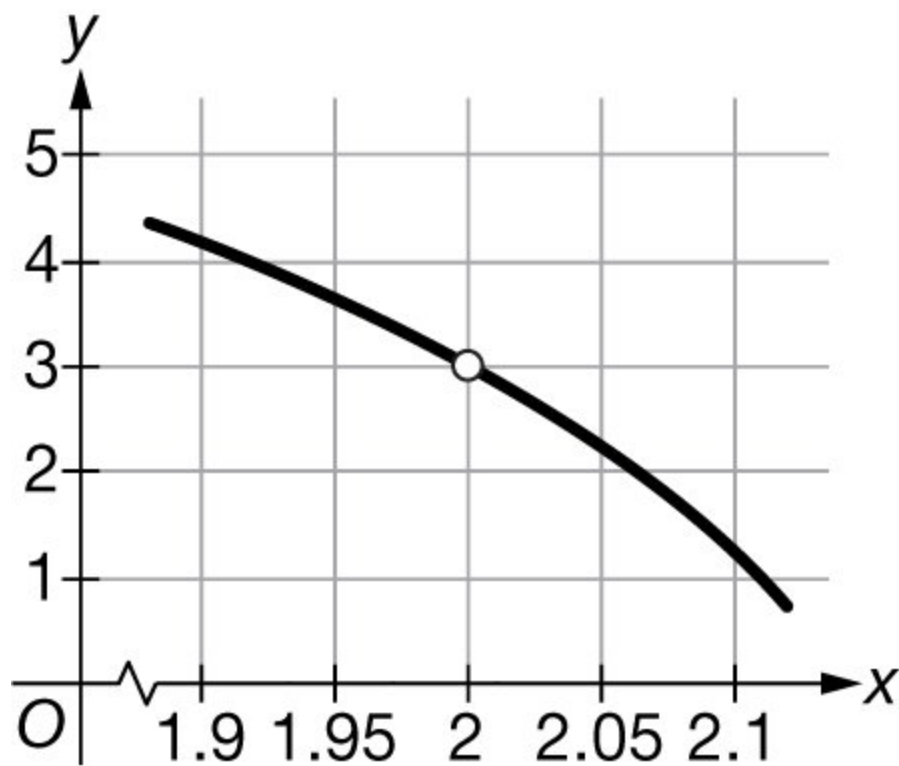
The table above gives selected values for a function f . Based on the data in the table, which of the following could not be the graph of f on the interval $1.9 \leq x \leq 2.1$?

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(A)

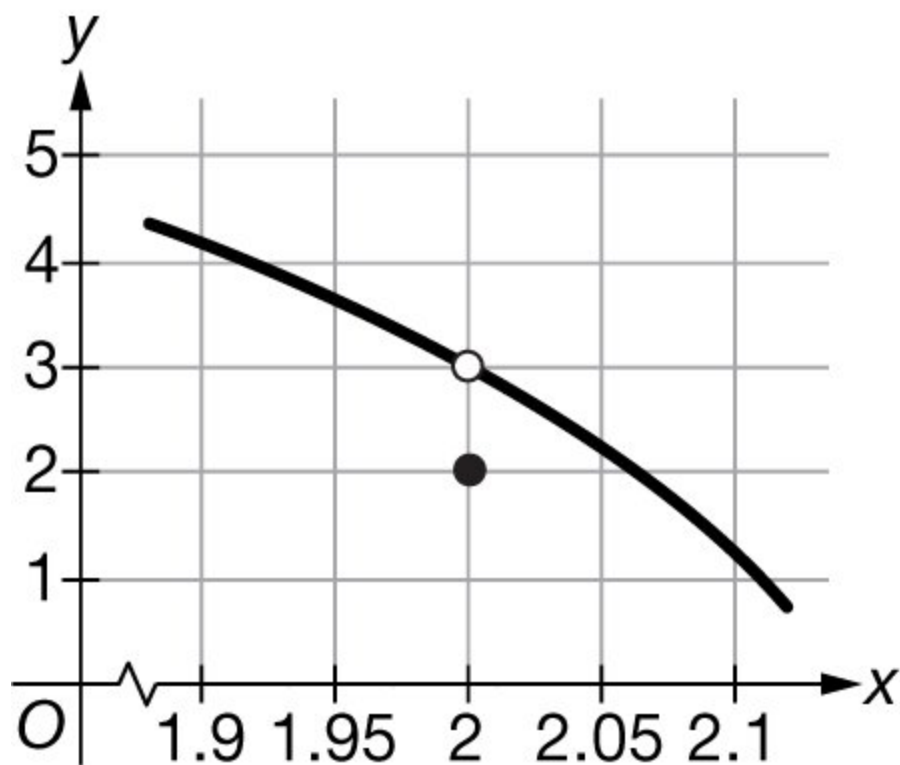


(B)

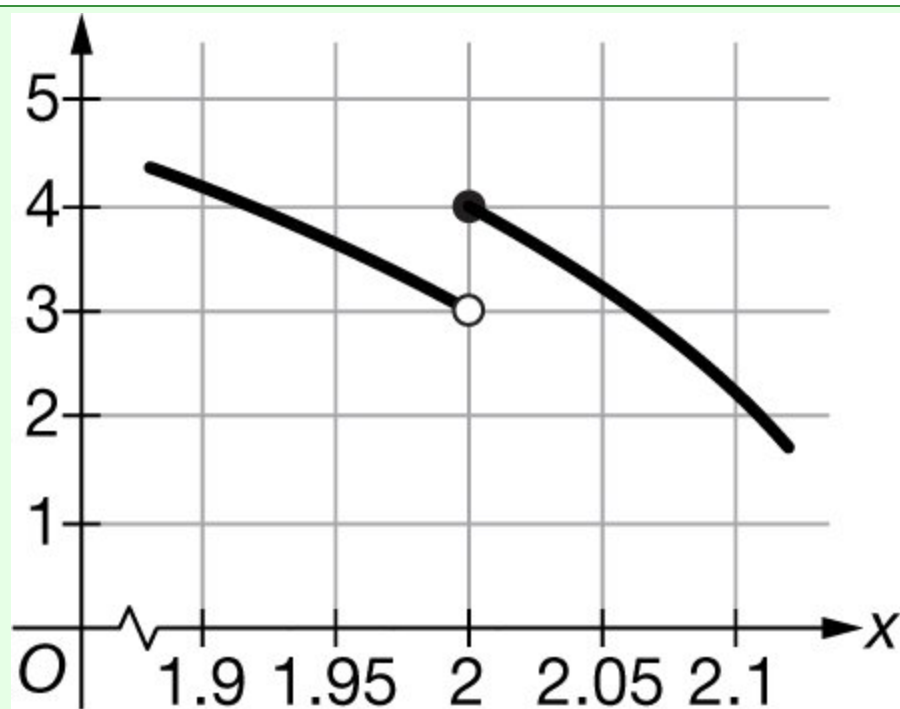


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(C)



(D)

**Answer D**

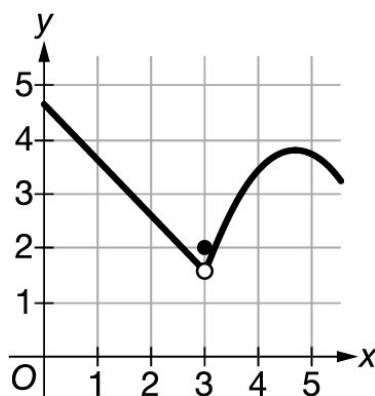
Correct. The data in the table suggest that $\lim_{x \rightarrow 2} f(x) = 3$. This function has a jump discontinuity at $x = 2$ where the

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limit does not exist. Therefore this could not be the graph of f .

95.

x	2.9	2.95	2.98	2.999	3.001	3.02	3.05	3.1
$f(x)$	1.7	1.65	1.62	1.601	1.603	1.66	1.747	1.89



The table above gives selected values for a function f . Also shown is a portion of the graph of f . The graph consists of a line segment for $x < 3$ and part of a parabola for $x > 3$. What is $\lim_{x \rightarrow 3} f(x)$?

- (A) 1.6 ✓
- (B) 1.602
- (C) 2
- (D) The limit does not exist.

Answer A

Correct. The graph of f indicates that the limit exists at $x = 3$ and has a value between 1 and 2. The linear segment in the graph of f has slope $\frac{1.650 - 1.700}{2.95 - 2.9} = \frac{-0.05}{0.05} = -1$. The equation of the line is $y = -(x - 2.9) + 1.7$. Therefore $\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} (-(x - 2.9) + 1.7) = -0.1 + 1.7 = 1.6$.

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96. Let f be the function given by $f(x) = \sqrt{x^4 - 16x^2}$.
- (a) Find the domain of f .
 - (b) Describe the symmetry, if any, of the graph of f .
 - (c) Find $f'(x)$.
 - (d) Find the slope of the line normal to the graph of f at $x = 5$.

Part A

1 point is earned for radicand ≥ 0

1 point is earned for $|x| \geq 4$

1 point is earned for $x = 0$

$$x^4 - 16x^2 \geq 0$$

$$x^2(x^2 - 16) \geq 0$$

$$x^2 \geq 16 \text{ or } x = 0$$

$$|x| \geq 4 \text{ or } x = 0$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for radicand ≥ 0

1 point is earned for $|x| \geq 4$

1 point is earned for $x = 0$

$$x^4 - 16x^2 \geq 0$$

$$x^2(x^2 - 16) \geq 0$$

$$x^2 \geq 16 \text{ or } x = 0$$

$$|x| \geq 4 \text{ or } x = 0$$

Part B

1 point is earned for the correct answer

Symmetric about the y -axis

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0	1
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The student response earns all of the following points:

1 point is earned for the correct answer

Symmetric about the y -axis

Part C

3 points are earned for correct derivative

$$f'(x) = \frac{1}{2}(x^4 - 16x^2)^{-\frac{1}{2}}(4x^3 - 32x)$$

$$= \frac{2x(x^2 - 8)}{|x|\sqrt{x^2 - 16}}$$



0	1	2	3
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The student response earns all of the following points:

3 points are earned for correct derivative

$$f'(x) = \frac{1}{2}(x^4 - 16x^2)^{-\frac{1}{2}}(4x^3 - 32x)$$

$$= \frac{2x(x^2 - 8)}{|x|\sqrt{x^2 - 16}}$$

Part D

1 point is earned for evaluating $f'(x)$ found in (c) at $x = 5$

1 point is earned for slope of normal

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$$f'(5) = \frac{2(5)(25-8)}{5\sqrt{25-16}}$$

$$= \frac{10(17)}{5\sqrt{9}} = \frac{170}{15} = \frac{34}{3}$$

$$\therefore \text{slope of normal line is } -\frac{3}{34}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for evaluating $f'(x)$ found in (c) at $x = 5$

1 point is earned for slope of normal

$$f'(5) = \frac{2(5)(25-8)}{5\sqrt{25-16}}$$

$$= \frac{10(17)}{5\sqrt{9}} = \frac{170}{15} = \frac{34}{3}$$

$$\therefore \text{slope of normal line is } -\frac{3}{34}$$